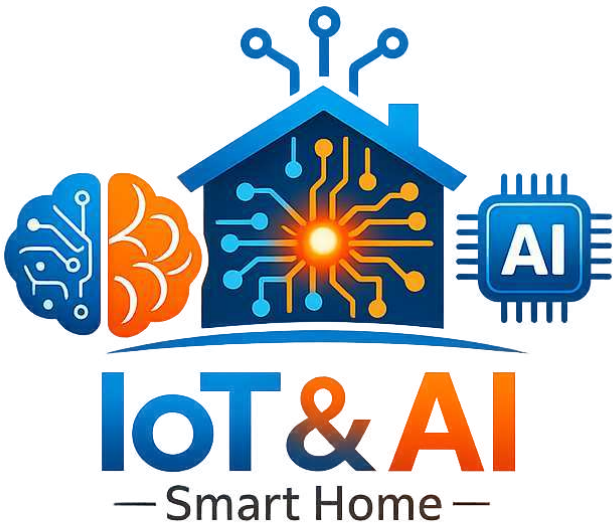
MSc AI Technology Portfolio Project

AIoT Smart Home ESP32-S3

AIoT Smart Home Simulation with Edge/TinyML Experimentation



Author: Alberto Reyes Castro

Technologies: ESP32-S3, Wokwi, Blynk, Edge Impulse / TinyML, Sensors

Focus Areas: AIoT, TinyML, Embedded AI, Smart Home Automation

Public portfolio version: October 2026 | **Technical code baseline:** `c96d98b`

Portfolio adaptation of an MSc Artificial Intelligence Technology project

Contents

1 Introduction **3**

2 Literature Review **3**
2.1 AI in IoT Systems . . . 3
2.2 Edge AI and Local Reasoning . . . 3
2.3 Smart Home Automation and Environmental Control . . . 4
2.4 Gas and Domestic Safety Monitoring . . . 4
2.5 Machine Learning for Embedded Safety Classification . . . 4
2.6 Research Gap . . . 4

3 System Design **4**
3.1 Overall House Architecture . . . 4
3.2 Common Control Philosophy . . . 5
3.3 Away Mode Security Logic . . . 5
3.4 Bedroom Module . . . 6
3.4.1 Design Rationale . . . 6
3.4.2 Hardware Structure . . . 6
3.4.3 Bedroom Behavioural Logic . . . 8
3.4.4 Bedroom Lighting and Blinds Logic . . . 8
3.4.5 Bedroom Thermal Logic . . . 11
3.5 Kitchen / Living Room Module . . . 13
3.5.1 Design Rationale . . . 13
3.5.2 Hardware Structure . . . 13
3.5.3 Kitchen/Living Behavioural Logic . . . 14
3.5.4 Gas and Environmental Safety Logic . . . 15
3.5.5 Kitchen/Living Thermal Logic . . . 16

4 AI Model Design, Evaluation & Deployment Path **18**
4.1 Role of AI in the Project . . . 18
4.2 Problem Type . . . 18
4.3 Dataset Collection and Labelling . . . 19
4.4 Impulse Design . . . 20
4.5 Training and Performance Metrics . . . 20
4.6 Deployment Strategy . . . 22

5 Implementation **22**
5.1 Simulation Environment . . . 22
5.2 Bedroom Module Implementation . . . 23
5.3 Kitchen/Living Room Implementation . . . 23
5.4 Blynk Mobile Dashboards and Remote Interaction . . . 25
5.5 Authentication and Security Logic . . . 27

6 Evaluation **28**

7 Ethical, Privacy & Security Considerations **28**

8 Discussion **29**

9 Conclusion & Future Work **29**

References **31**

A Appendix A: Additional Simulation Figures (Wokwi) **32**

B Appendix B: Public Repository and Supporting Material **33**

C Appendix C: Brief User Guide **33**
C.1 Purpose 33
C.2 Required Platforms and Materials 33

1 Introduction

Artificial Intelligence of Things (AIoT) is an important area within embedded systems, smart environments, and edge computing. Rather than only connecting sensors and actuators, AIoT combines sensing, local reasoning, and real-time response to create context-aware behaviour (Al-Fuqaha et al., 2015; Shi et al., 2016). Smart homes are one of its clearest applications because they bring together comfort, privacy, safety, energy management, and user-focused automation (Stojkoska and Trivodaliev, 2017; Ray, 2018).

This project develops a modular smart home prototype using ESP32-S3 room controllers. Each room works as an independent embedded module with its own sensors, actuators, local logic, and remote monitoring interface, while still forming part of a wider house architecture. This room-level approach improves modularity, explainability, and scalability without relying on a single central controller.

The prototype focuses on a bedroom and a kitchen/living room. Both follow the same overall design philosophy but serve different priorities. The bedroom emphasises comfort, privacy, rest, and support for vulnerable users, whereas the kitchen/living room prioritises functional control, environmental monitoring, gas safety, and emergency response. This reflects a core principle of the project: smart-home behaviour should adapt to the role of each domestic space rather than apply identical logic everywhere.

Technically, the project combines low-cost embedded sensing, explainable rule-based control, optional Blynk-based IoT interaction, and an Edge Impulse workflow. The bedroom remains mainly rule-based because its behaviour can be expressed through clear comfort and privacy rules. The kitchen/living room also includes a historical Edge Impulse environmental classifier trained using gas score, temperature, and humidity; the hardened public Wokwi build keeps that generated runtime disabled by default. The model predicts five states: `NORMAL`, `COOKING_FUMES`, `GAS_DANGER`, `HEAT_ANOMALY`, and `GAS_HEAT_EMERGENCY`.

The main aim is to design and prototype a modular AIoT smart home that supports local decisions for comfort, privacy, safety, and security. To achieve this, the project develops room-specific embedded logic, integrates sensors and actuators, evaluates behaviour through simulation, and demonstrates an initial embedded machine-learning workflow for environmental safety classification in the kitchen/living room.

2 Literature Review

2.1 AI in IoT Systems

AIoT extends the Internet of Things by moving part of the decision process closer to the device. In conventional IoT systems, sensors often send data elsewhere for processing; in AIoT, more reasoning can happen locally. In domestic environments, this is valuable because latency, connectivity dependence, and privacy concerns can reduce the usefulness of a smart system. The literature shows applications in occupancy detection, environmental control, anomaly monitoring, adaptive lighting, and assisted living. However, the degree of intelligence varies widely, from threshold-based automation to full cloud or edge machine-learning pipelines (Al-Fuqaha et al., 2015; Ray, 2018).

2.2 Edge AI and Local Reasoning

A key distinction is between cloud-centred and edge-centred intelligence. Cloud-based AI supports stronger remote computation and easier large-scale management, but can introduce latency, connectivity dependence, and privacy concerns. In domestic systems, responses to motion, comfort, privacy, or gas-related events are often most useful when they occur locally and immediately. For this reason, edge AI has become increasingly important in smart environments. Even when a full learned model is not deployed, the same principle still applies: room-level local reasoning can make the system faster, more resilient, and easier to trust (Shi et al., 2016).

Recent reviews of edge AI and edge machine learning reinforce this design choice. Edge AI is increasingly used to move inference towards resource-constrained devices, reducing latency and cloud dependence while improving privacy and resilience (Merenda et al., 2020; Singh and Gill, 2023). In the IoT context, TinyML is also presented as a practical way to run lightweight models close to the sensor source, which matches the ESP32-S3 and Edge Impulse direction of this project (Dutta and Bharali, 2021).

2.3 Smart Home Automation and Environmental Control

Smart home literature often treats lighting, thermal control, assisted care, and security as separate functions. Lighting is commonly based on occupancy or brightness thresholds, thermal control on temperature and humidity ranges, and security on simple intrusion alarms. These approaches are useful but can become limited when context is ignored. A stronger design should allow the same input to produce different responses depending on room purpose, time, and user profile. This supports the modular room-specific structure used in this project (Stojkoska and Trivodaliev, 2017).

2.4 Gas and Domestic Safety Monitoring

Gas and smoke sensing is especially relevant in kitchen environments. However, low-cost gas sensors do not always separate harmless vapour, cooking fumes, and higher gas-score conditions clearly. The practical design challenge is therefore to remain sensitive to risk without becoming too reactive during everyday use. A common response is to combine thresholds with persistence over time, contextual interpretation, and secondary environmental signals. This directly informs the gas-safety logic used in the kitchen/living room module.

2.5 Machine Learning for Embedded Safety Classification

TinyML and embedded machine learning are useful when a system must classify patterns from several sensor values without sending data to a remote service. However, safety-related machine learning must be introduced carefully. A learned model should not replace deterministic protection logic unless it has been tested thoroughly in the target environment. For this project, the hybrid-capable architecture therefore keeps rule-based gas escalation as the deterministic safety path. The optional classifier is deliberately small and MCU-oriented, consistent with TinyML constraints around memory, latency, energy use, and local inference (Warden and Situnayake, 2019; Merenda et al., 2020; Dutta and Bharali, 2021).

2.6 Research Gap

Many smart-home projects remain either too generic or too centralised. They often apply similar logic across all rooms, rely heavily on cloud services, or prioritise novelty over explainability. There is therefore value in a modular edge-oriented architecture where each room keeps its own priorities while still contributing to a coherent house-wide design. This project addresses that gap through two differentiated room modules built around a shared control philosophy.

3 System Design

3.1 Overall House Architecture

The proposed smart home uses a modular architecture in which each room is represented by an ESP32-S3 controller. Each controller gathers room-specific sensor data, interprets the local context, and applies local decision logic without depending on a permanent cloud service. The current prototype includes a bedroom module and a kitchen/living room module. They are implemented as separate subsystems but share common ideas: operating modes, user profiles, occupancy awareness, local alerts, and safety overrides. This creates house-level consistency while still allowing each room to behave differently according to its function.

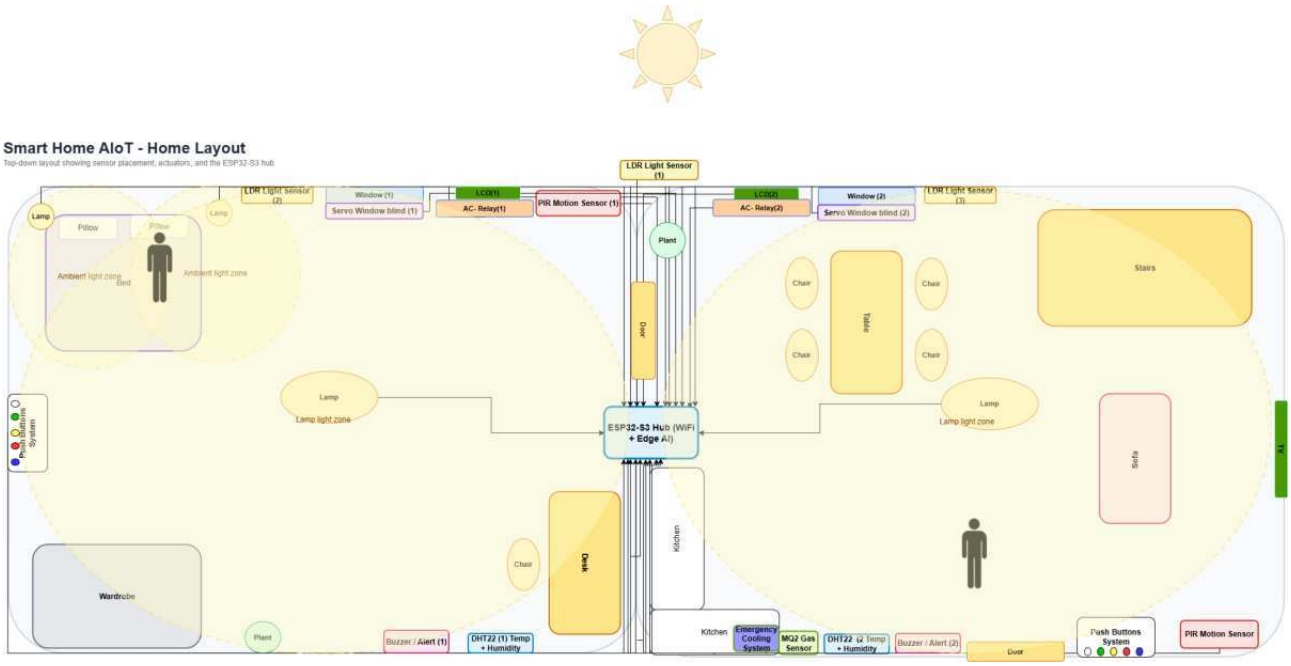


Figure 1: Overall house architecture and room distribution designed in draw.io.



(a) 2D plan view.

(b) 3D plan view.

Figure 2: House visualisation created in Planner 5D, showing the 2D plan and the 3D view.

3.2 Common Control Philosophy

The house follows a shared logic based on three common room modes (*Normal*, *Privacy*, and *Eco*), three user profiles (*Standard*, *Elderly*, and *Child*), and a wider *Home/Away* house state. *Sleep* is kept only for the bedroom because it is primarily relevant to rest-oriented behaviour. This framework gives the system consistency while still allowing room-specific responses.

3.3 Away Mode Security Logic

Unlike the comfort-oriented room modes, *Away* is treated as a house security state. When the system is armed, comfort behaviour is reduced and the priority becomes monitoring, intrusion escalation, authenticated recovery, and energy saving. Motion is evaluated within a security window so that the system can distinguish between passive monitoring, suspicious motion, and confirmed intruder alert. The design also includes pet-aware behaviour to reduce false alarms caused by small domestic movement.



AWAY Mode – Security Behaviour (General Logic)

Decision table for HOME vs AWAY security behaviour

Purpose: protect the home when no authorized users are present

Sensors used: PIR motion sensor, Door/Window sensor, Camera (optional), System status

Condition	Bedroom Response	Kitchen / Living Response	Why this behaviour?
AWAY mode + no motion detected	• Blinds: CLOSED • Ambient light: OFF • AC: OFF • System status: Armed	• Blinds: CLOSED • Ambient light: OFF • AC: OFF • System status: Armed	Mantains the house in a secure state, minimizing energy use and preventing unauthorized access.
AWAY mode + first motion detected	• Keep blinds: CLOSED • Ambient light: OFF • AC: OFF • Record/Log event	• Keep blinds: CLOSED • Ambient light: OFF • AC: OFF • Record/Log event	A single motion can be normal (e.g., pet). The system monitors before escalating.
AWAY mode + repeated motion (suspicious)	• Keep blinds: CLOSED • Ambient light: OFF • AC: OFF • Trigger intruder alert • Notify user	• Keep blinds: CLOSED • Ambient light: FLASH / ON • AC: OFF • Trigger intruder alert • Notify user	Repeated motion suggests intrusion. The system raises an alert and notifies the user.
AWAY mode + pet mode enabled	• Keep blinds: CLOSED • Ambient light: OFF • AC: OFF • Ignore small/pet motion	• Keep blinds: CLOSED • Ambient light: OFF • AC: OFF • Ignore small/pet motion	Reduces false alarms caused by pet movement.
AWAY mode + intruder alert active	• Blinds: CLOSED • Ambient light: ON (for visibility) • AC: OFF • Sound alarm (if available) • Continuous notifications	• Blinds: CLOSED • Ambient light: ON (for visibility) • AC: OFF • Sound alarm (if available) • Continuous notifications	Maximizes deterrence and user awareness until the situation is resolved.
AWAY mode + correct authentication	• Return to previous mode (Normal / Sleep / Privacy / Eco) • Blinds, lights, AC restored according to that mode • System status: Disarmed	• Return to previous mode (Normal / Sleep / Privacy / Eco) • Blinds, lights, AC restored according to that mode • System status: Disarmed	The system returns to normal operation once the user is authenticated.

AWAY Mode – AC Behaviour (All Profiles)

Decision table for bedroom AC control in AWAY mode

Purpose: minimize energy consumption and avoid unnecessary cooling

AC behaviour in AWAY mode: AC is kept **OFF** for all profiles (Standard, Elderly, Child) regardless of temperature or humidity.

Scenario	Standard	Elderly	Child	Why this configuration?
AWAY mode + any temperature / humidity	AC OFF	AC OFF	AC OFF	When the house is empty, the AC remains OFF to save energy and reduce wear. Comfort is restored when the user returns and selects the desired mode.

Note: When the system leavees AWAY mode (correct authentication), AC behaviour follows the selected room mode (Normal, Sleep, Privacy or Eco) and the corresponding profile thresholds.

Figure 3: Away mode security logic across the implemented smart room modules. As shown in this figure, Away mode works as a protective state above the normal room modes. In this state, blinds remain in a defensive position, ordinary comfort functions are reduced, and repeated motion can escalate from monitoring to suspicious motion and then to intruder alert. After authentication, the system returns to Home state instead of staying armed, which helps avoid a repeated alarm loop.

3.4 Bedroom Module

3.4.1 Design Rationale

The bedroom is designed around comfort, privacy, rest, and support for vulnerable users. It therefore includes adaptive blinds, ambient support lighting, a manual lamp state, thermal monitoring, AC output, and occupancy-aware logic. Compared with the kitchen/living room, it requires finer behavioural distinctions, especially at night and across user profiles.

3.4.2 Hardware Structure

The bedroom module includes an ESP32-S3, PIR motion sensor, indoor and outdoor LDR sensors, a servo-controlled blind, smart ambient light output, manual lamp state indicator, DHT22 temperature and humidity sensor (Aosong Electronics, n.d.), AC relay, buzzer, control buttons, and an LCD interface.

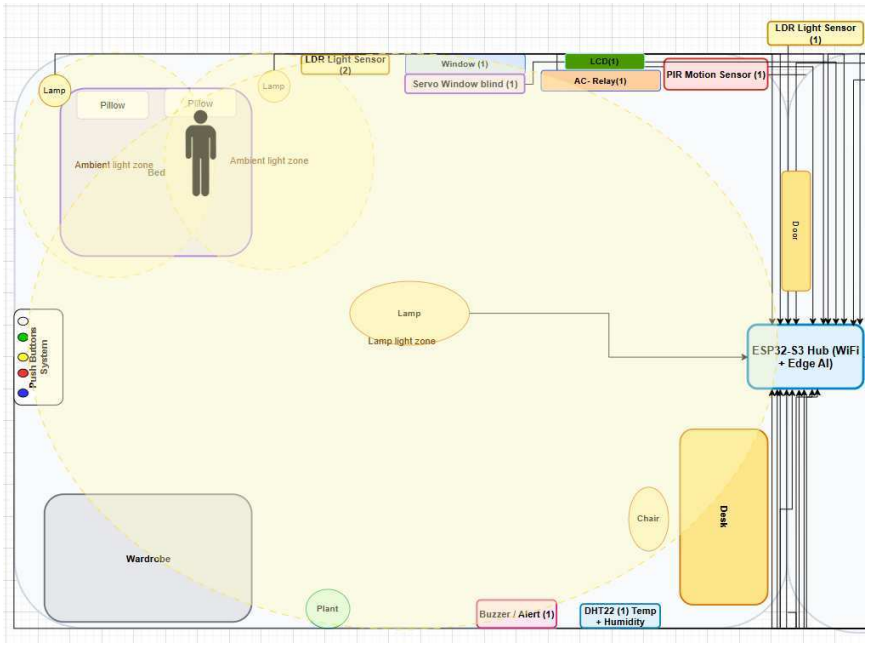


Figure 4: Bedroom hardware layout and logical arrangement in draw.io.



(a) 2D plan view.



(b) 3D view 1.



(c) 3D view 2.



(d) 3D view 3.

Figure 5: Bedroom visualisation created in Planner 5D, including the 2D plan and three-dimensional views.

3.4.3 Bedroom Behavioural Logic

The bedroom uses four modes. *Normal* supports balanced everyday behaviour, *Sleep* prioritises rest, *Privacy* restricts visibility and daylight exposure, and *Eco* reduces unnecessary energy use while keeping basic support available. Profiles refine this behaviour further: *Standard* provides the default response, *Elderly* extends presence hold time and favours gentler night support, and *Child* adds similarly protective behaviour together with protected deactivation through PIN logic.

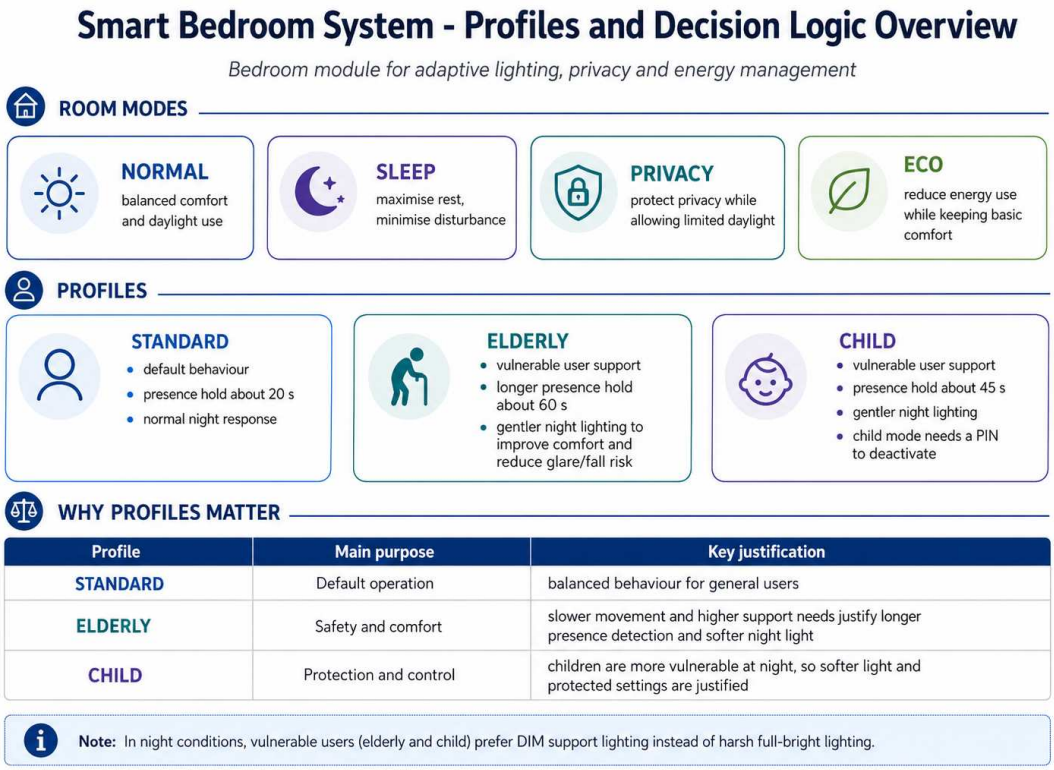


Figure 6: Bedroom profiles and overall control logic overview.

3.4.4 Bedroom Lighting and Blinds Logic

Lighting and blind behaviour are central to the bedroom design. *Sleep* mode keeps the room as dark as possible, while the *Elderly* and *Child* profiles still allow limited night support in selected motion situations. The detailed decision tables below summarise the designed responses.

NORMAL Mode - Behaviour by Profile

Decision table for the smart bedroom in NORMAL mode

Purpose: balanced comfort and daylight use

Presence hold: Standard 20 s | Elderly 60 s | Child 45 s

Scenario	Standard	Elderly	Child	Why this configuration?
Day / transition + no motion	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	Blinds maximise useful daylight during the day. With no motion, the system keeps ambient light OFF to save energy and maintain natural comfort.
Day / transition + motion detected	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	Daylight is prioritised. When the room is still dark and someone is present, DIM ambient light supports visibility while preserving daylight and comfort.
Night + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	At night, blinds stay closed for privacy and rest. With no motion, ambient light remains OFF to avoid unnecessary light and save energy.
Night + motion detected	• Blinds: CLOSED • If room is dark: ambient light ON • If room is medium: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM • If room is medium: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM • If room is medium: ambient light DIM	Night-time visibility is supported based on user profile. Elderly and child profiles use gentler light to reduce glare and improve comfort and safety.

Manual lamp override: if the manual lamp is ON, the automatic ambient light stays OFF.

Figure 7: Bedroom Normal mode decision table for lighting and blinds.

SLEEP Mode - Behaviour by Profile

Decision table for the smart bedroom in SLEEP mode

Purpose: maximise rest and minimise disturbance

Presence hold: Standard 20 s | Elderly 60 s | Child 45 s

Scenario	Standard	Elderly	Child	Why this configuration?
Day / transition + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	Sleep mode prioritises rest during the day/transition period. With no motion, the room remains quiet and dark.
Day / transition + motion detected	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	Even if motion is detected, sleep mode suppresses daytime lighting support so rest remains the main priority.
Night + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	At night with no motion, the room stays fully dark to maximise rest and avoid unnecessary light.
Night + motion detected	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • If room is dark: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM	Vulnerable users receive DIM support lighting at night to improve safety and reassurance, while Standard profile keeps lighting OFF to minimise disturbance.

Note: Sleep mode gives maximum priority to rest. Vulnerable users may still receive DIM night support for safety.

Figure 8: Bedroom Sleep mode decision table for lighting and blinds.

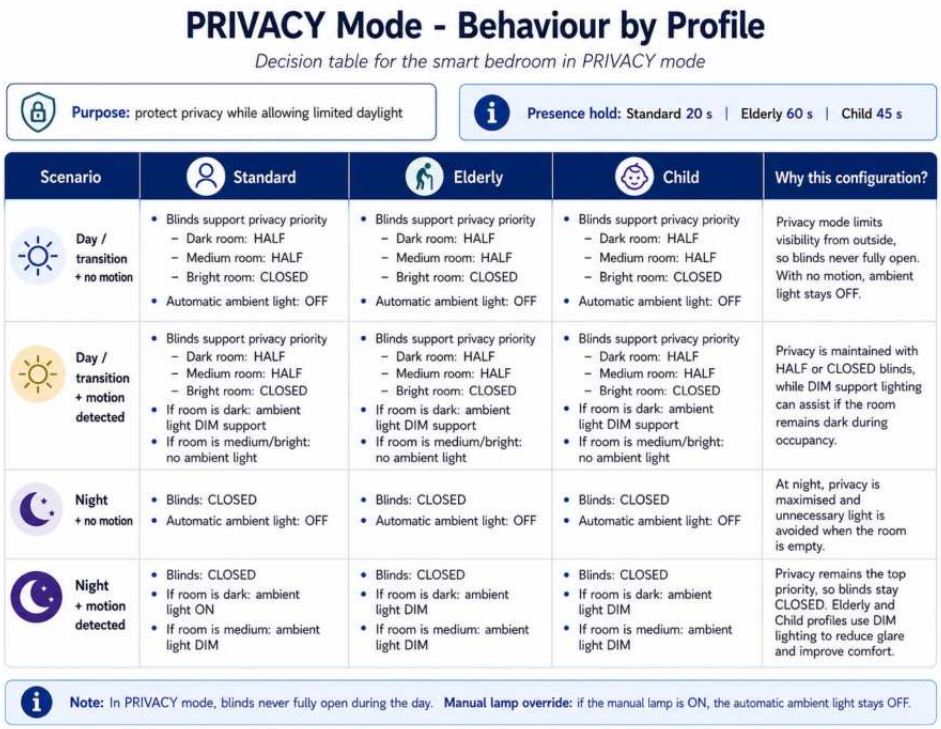


Figure 9: Bedroom Privacy mode decision table for lighting and blinds. In Privacy mode, half-open blinds allow natural light to enter the room while still preventing visibility from outside. This aims to preserve useful indoor illumination while reducing visibility from outside; the simulation does not quantify real-world privacy performance.

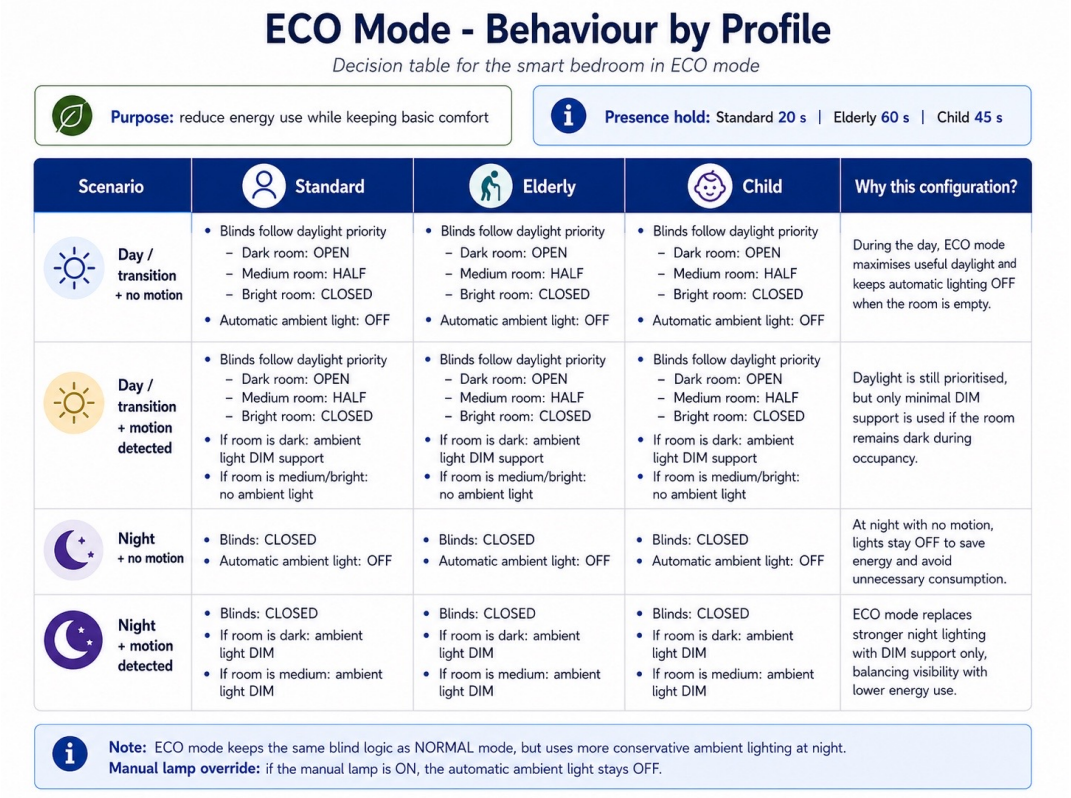


Figure 10: Bedroom Eco mode decision table for lighting and blinds.

3.4.5 Bedroom Thermal Logic

Bedroom thermal comfort is managed through DHT22 sensing and AC relay control. The response varies by profile and mode: vulnerable user profiles justify slightly earlier intervention, while *Eco* delays activation to reduce energy use. The following figures summarise the designed AC behaviour.

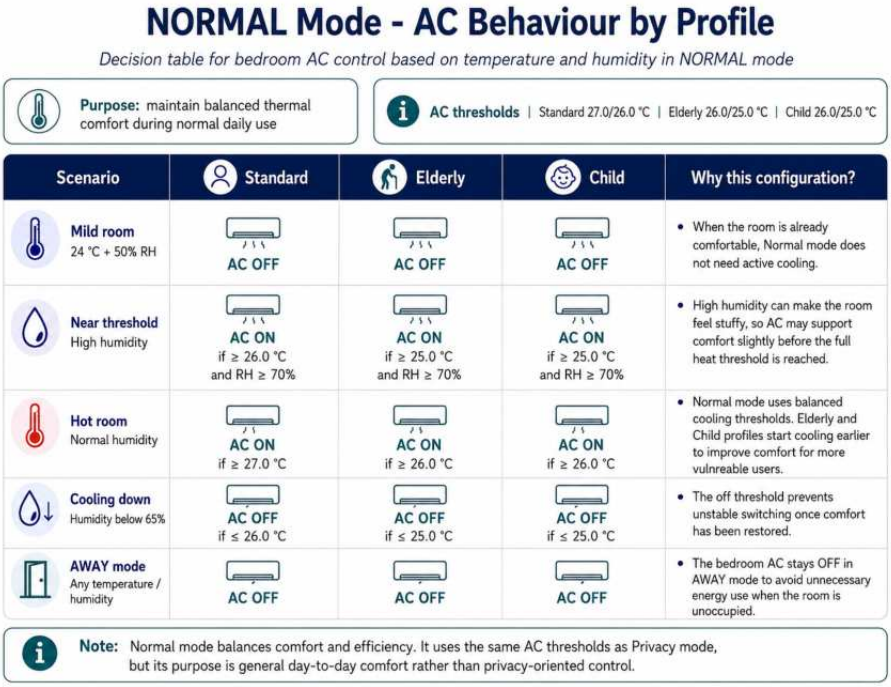


Figure 11: Bedroom Normal mode AC behaviour by profile.

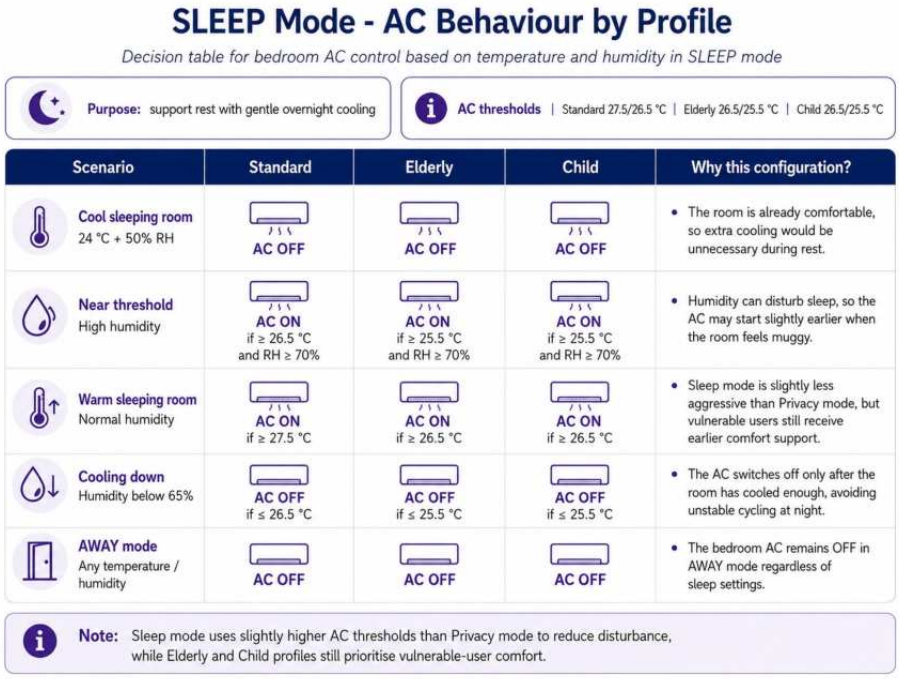


Figure 12: Bedroom Sleep mode AC behaviour by profile.

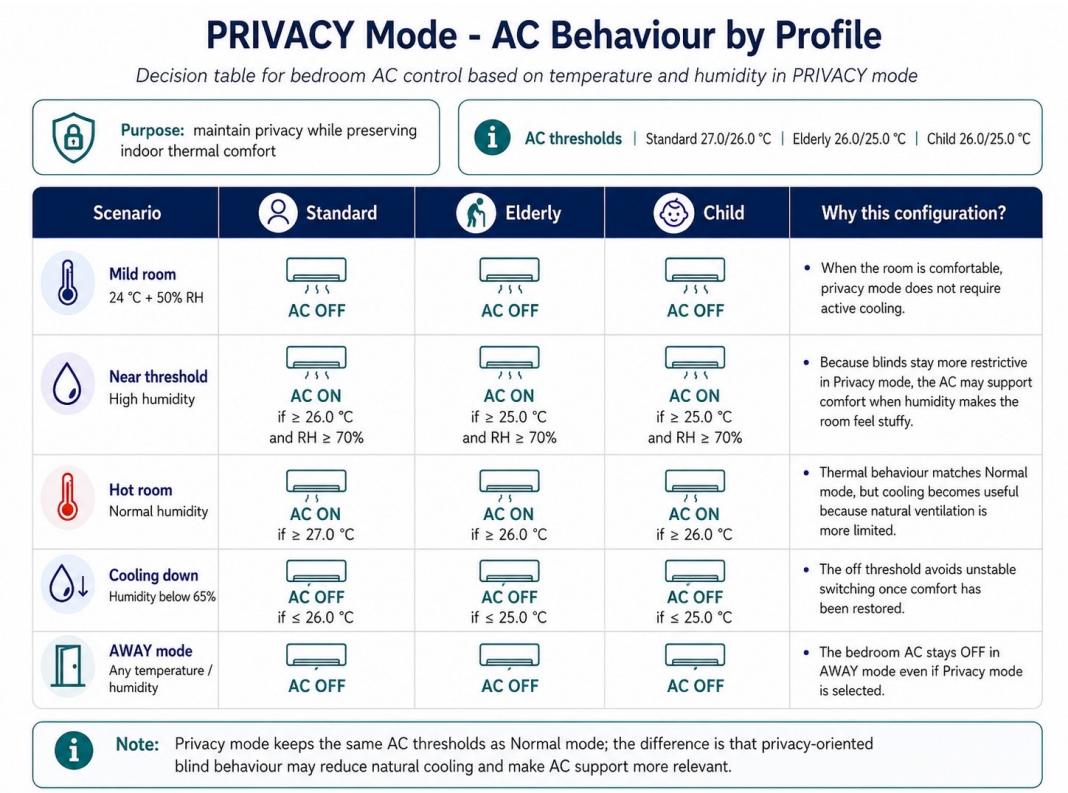


Figure 13: Bedroom Privacy mode AC behaviour by profile.

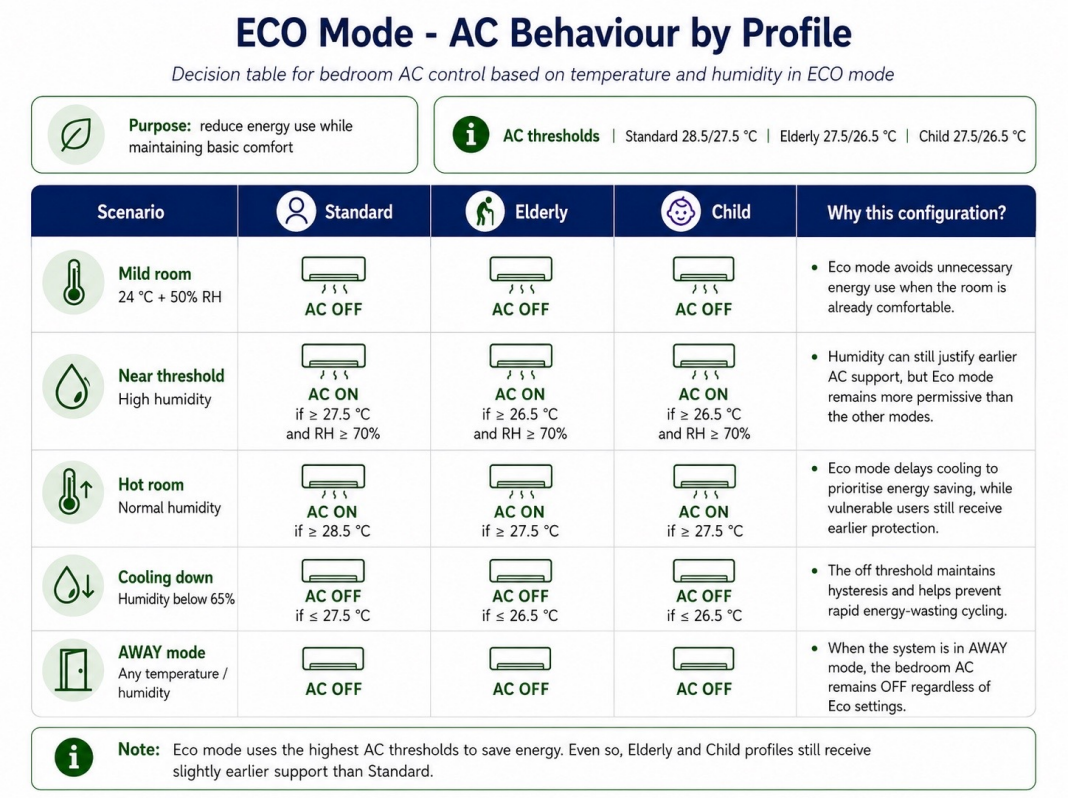


Figure 14: Bedroom Eco mode AC behaviour by profile.

3.5 Kitchen / Living Room Module

3.5.1 Design Rationale

The kitchen/living room follows the same general philosophy as the bedroom but serves a different role. It is a shared, active, and safety-sensitive environment. For this reason, the design avoids bedroom-style automatic ambient support lighting and instead uses a single manually controlled main lamp. The main priorities are functional control, thermal comfort, occupancy awareness, gas safety monitoring, and emergency response.

This module is also the strongest candidate for machine-learning experimentation because it combines several environmental signals that can describe risk together. Unlike the bedroom, where most behaviour can be explained through comfort and privacy rules, the kitchen/living room pairs deterministic safety control with an optional Edge Impulse classifier used as advisory telemetry.

3.5.2 Hardware Structure

The kitchen/living room module includes an ESP32-S3, PIR motion sensor, indoor and outdoor LDR sensors, MQ2 gas sensor (Zhengzhou Winsen Electronics Technology Co., Ltd., 2021), DHT22 temperature and humidity sensor (Aosong Electronics, n.d.), servo-controlled blinds, manual main lamp output, AC relay, buzzer, LCD display, sprinkler or emergency-cooling indicator LED, and buttons for mode, profile, house state, and child mode interaction.

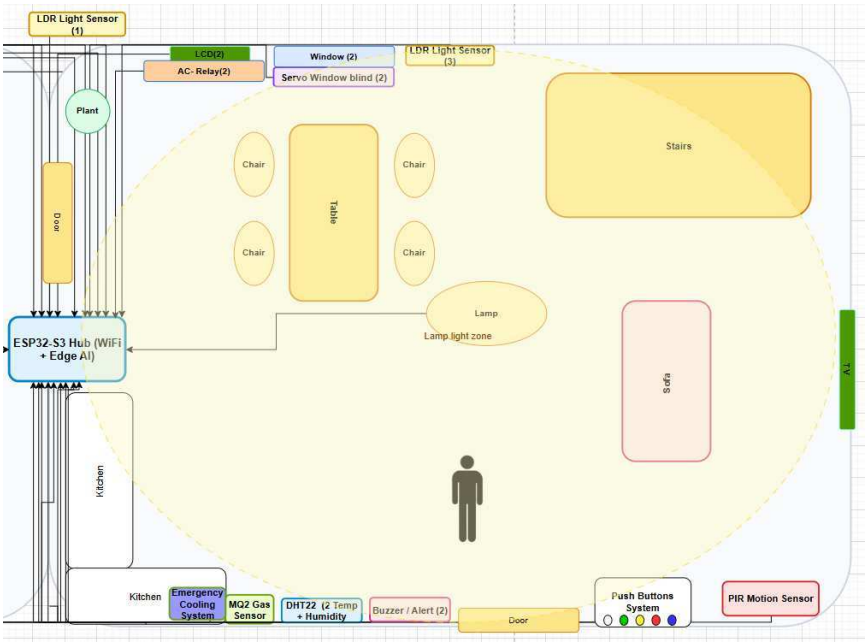


Figure 15: Kitchen/living room hardware layout and logical arrangement in draw.io.



(a) 2D plan view.



(b) 3D view 1.



(c) 3D view 2.



(d) 3D view 3.

Figure 16: Kitchen/living room visualisation created in Planner 5D, including the 2D plan and three-dimensional views.

3.5.3 Kitchen/Living Behavioural Logic

The final kitchen/living room design uses three room modes: *Normal*, *Privacy*, and *Eco*. *Sleep* was removed because it added little value in a shared functional room. The module still keeps the same profile logic as the rest of the house: *Standard* gives the default response, while *Elderly* and *Child* apply slightly more cautious thresholds and longer response timings. The room also supports the global *Home/Away* state, so repeated motion while armed can escalate from suspicious motion to intruder alert.

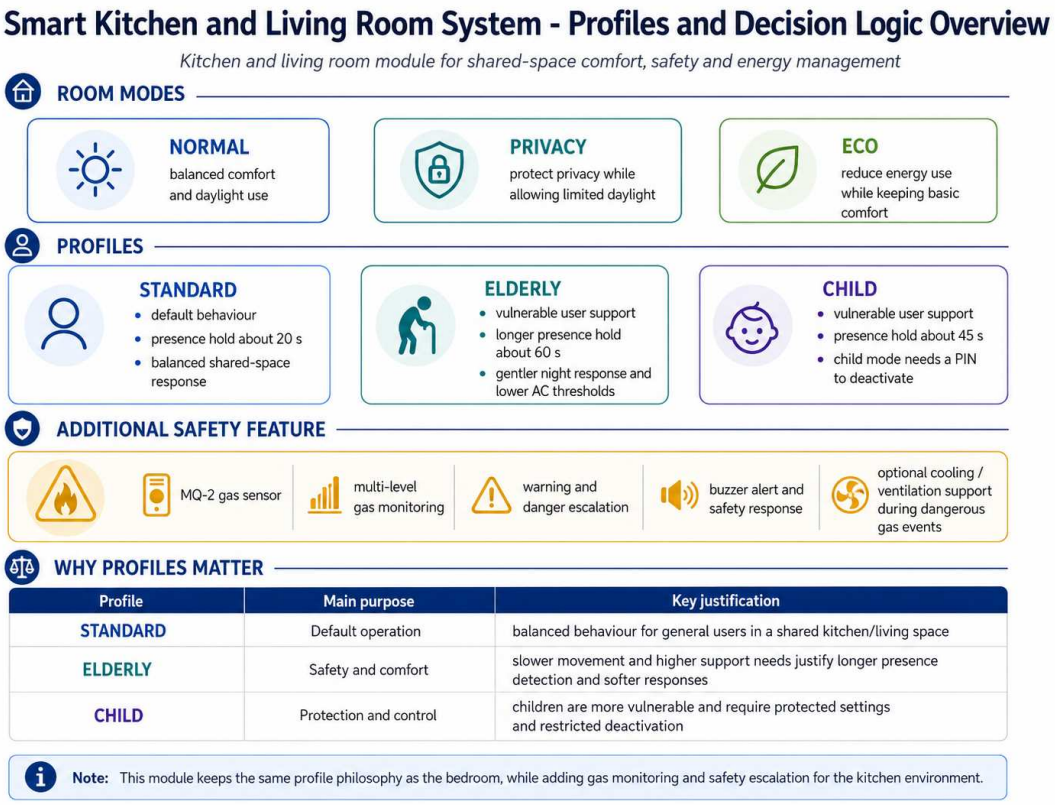


Figure 17: Kitchen/living room system overview. This shared space module follows the same overall house logic used in the bedroom, including the *Normal*, *Privacy*, and *Eco* modes, together with the *Standard*, *Elderly*, and *Child* profiles. However, *Sleep* mode is not implemented in this area, as the kitchen/living room is not mainly intended for night time rest. Unlike the bedroom, this module does not use automatic ambient lighting and relies on a manually controlled main lamp. Alongside blind and thermal control, the module integrates gas monitoring, heat anomaly detection, and emergency cooling response.

3.5.4 Gas and Environmental Safety Logic

The kitchen/living room adds the main safety dimension of the house: gas and environmental risk detection. Because a low-cost MQ2 sensor cannot cleanly separate harmless vapour, cooking fumes, and dangerous events, the system does not depend on one instant reading. This interpretation is consistent with using the MQ2 as a broad combustible-gas sensing element rather than as a precise domestic gas analyser (Zhengzhou Winsen Electronics Technology Co., Ltd., 2021). Instead, the actionable rule-based state uses thresholds, persistence timers, hysteresis, and staged escalation. A historical Edge Impulse experiment adds an optional classifier using three input features:



Figure 18: Input features.

The model predicts five environmental safety states:

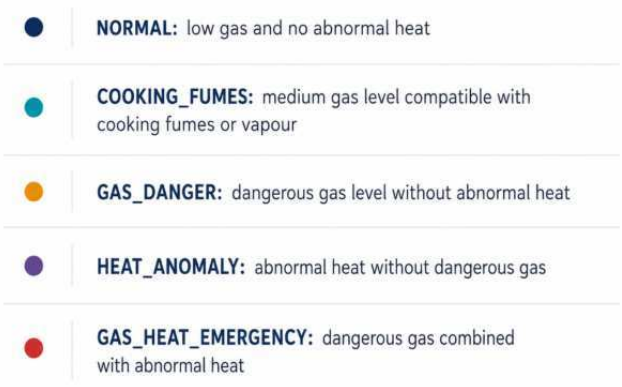


Figure 19: Five trained target classes. The historical Edge Impulse model-testing view also displays an `UNCERTAIN` outcome bucket; it is not presented here as a sixth trained class.

This gives temperature a clearer role in the logic: temperature above the project threshold without a high gas-score condition becomes `HEAT_ANOMALY`, while a high gas score combined with high temperature becomes `GAS_HEAT_EMERGENCY`. The latter activates the emergency-cooling response represented by a blinking sprinkler LED. The result is a hybrid-capable structure in which the gas state machine remains authoritative, while genuine model inference can be enabled separately after supplying and validating a generated library.

3.5.5 Kitchen/Living Thermal Logic

The kitchen/living room also uses DHT22-based thermal monitoring and AC control. This logic is simpler than in the bedroom because the main focus is shared-space comfort and safety rather than rest support. *Eco* delays AC intervention to reduce energy use, while the *Elderly* and *Child* profiles justify slightly more cautious thresholds. Safety overrides take priority: when cooking fumes or gas danger are active, the AC is switched off so that potentially contaminated air is not circulated through the room.

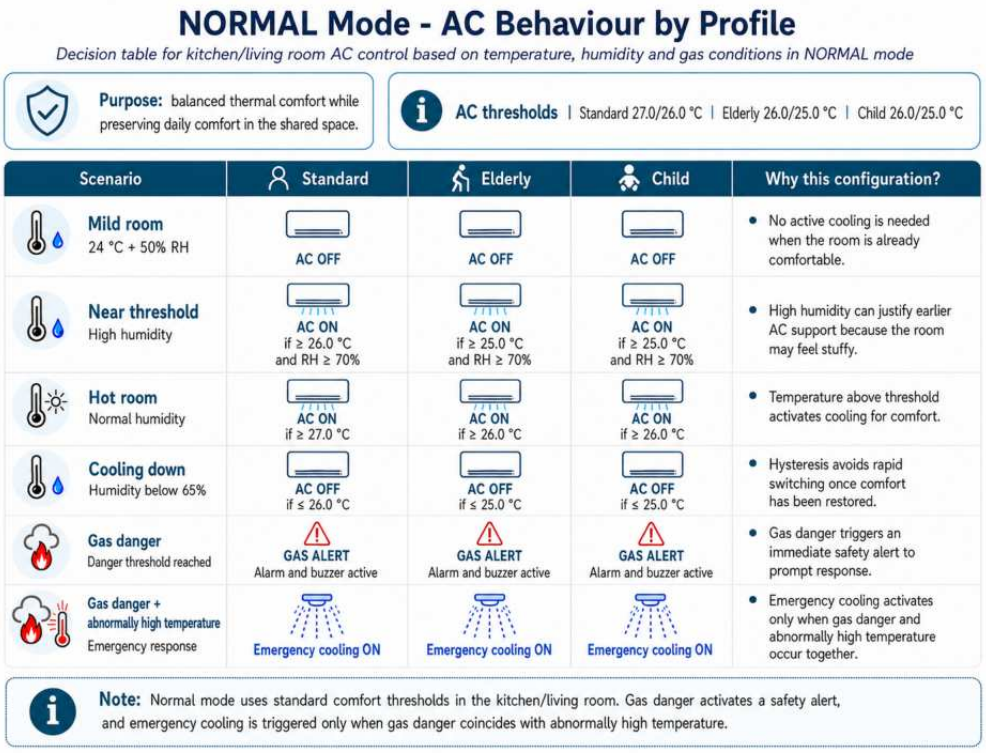


Figure 20: Kitchen/living room Normal mode AC behaviour by profile.

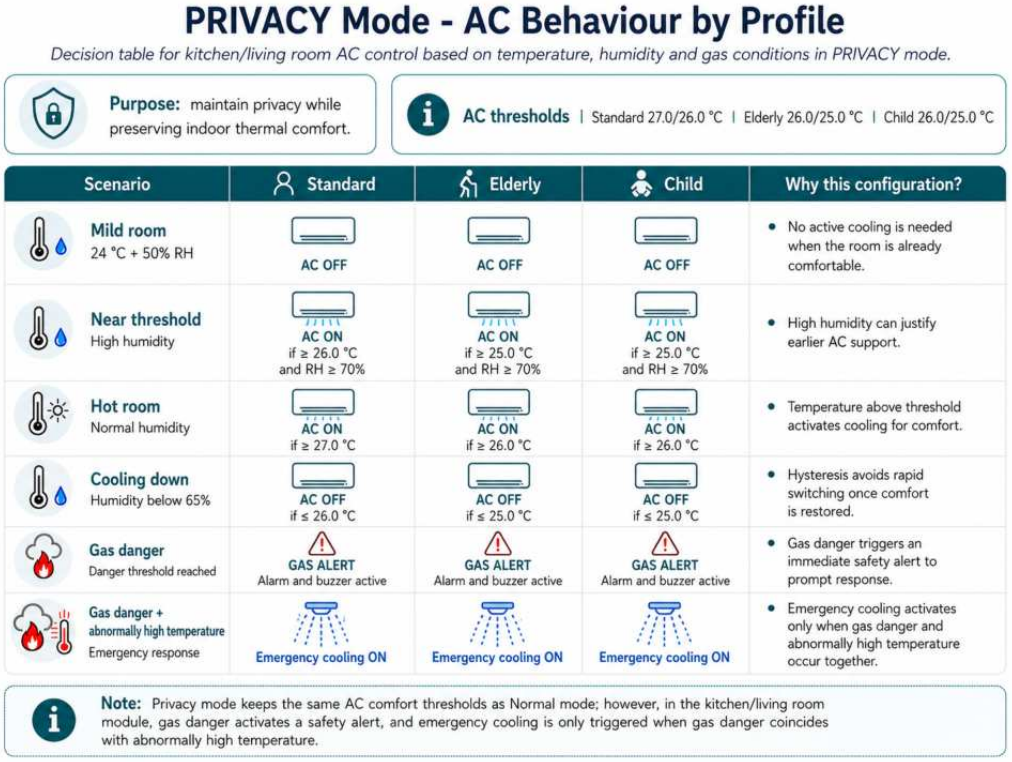


Figure 21: Kitchen/living room Privacy mode AC behaviour by profile.

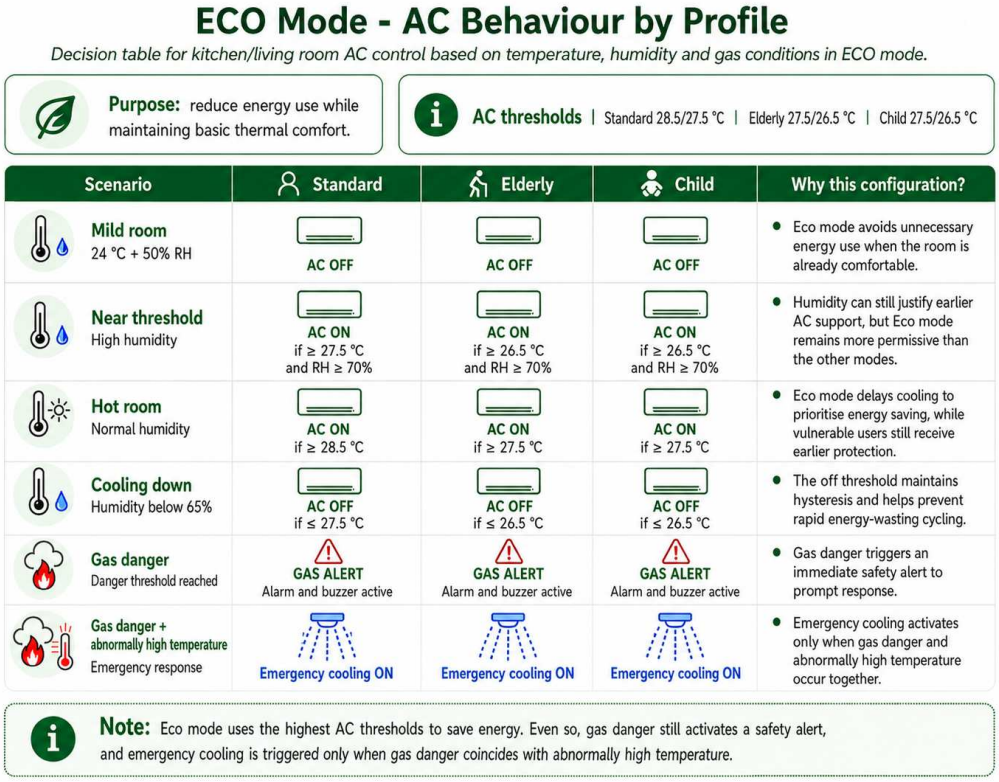


Figure 22: Kitchen/living room Eco mode AC behaviour by profile.

4 AI Model Design, Evaluation & Deployment Path

The AI part of the project is deliberately limited to the most safety-relevant and sensor-rich subsystem: environmental-state classification in the kitchen/living room. This keeps the task focused, explainable, and suitable for embedded experimentation with an optional deployment path.

4.1 Role of AI in the Project

The AI component is not intended to replace or control the rule-based safety system. The deterministic state machine remains the sole authority for safety actuators because its timing and escalation behaviour are explicit and testable. In the historical model experiment, Edge Impulse converts the environmental logic into a compact classifier based on `gas_score`, `temperature`, and `humidity` (Edge Impulse, n.d.a). The public repository preserves this workflow and its integration path, but disables generated-model inference by default.

The workflow covers dataset creation, labelling, model training, testing, performance evaluation, and historical Arduino-library export. Because the dataset is simulated and rule-labelled from the same designed boundaries used by the rules, strong model performance is expected and does not establish independent real-world validity. Deterministic rules therefore remain the default safety path.

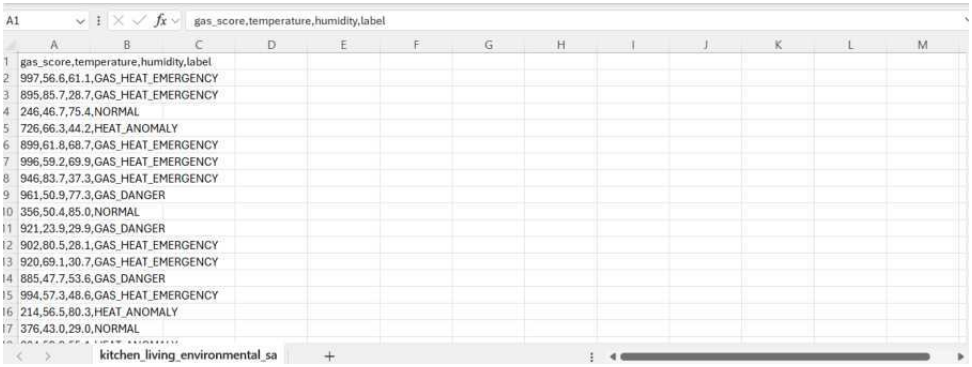
4.2 Problem Type

The selected problem is a multi-class classification task using the three numerical inputs shown in Figure 18 and the five trained target classes in Figure 19. The historical Edge Impulse model-testing interface additionally displays an `UNCERTAIN` outcome bucket; it is not treated here as a sixth trained class. This framing preserves a direct link between environmental state and embedded response without claiming chemical identification from the MQ2 signal.

4.3 Dataset Collection and Labelling

A custom dataset was created from the kitchen/living room simulation in Wokwi using the same ESP32-S3, MQ2, and DHT22 setup as the implemented module. This was preferable to a public dataset because the target classes were specific to the environmental safety states defined for this project. Data were collected through a simplified ESP32-S3 sketch that removed Blynk, LCD logic, buttons, and actuator control and printed clean CSV rows in the form:

```
gas_score,temperature,humidity,label
```



	gas_score,temperature,humidity,label
1	gas_score,temperature,humidity,label
2	997,56.6,61.1,GAS_HEAT_EMERGENCY
3	895,85.7,28.7,GAS_HEAT_EMERGENCY
4	246,46.7,75.4,NORMAL
5	726,66.3,44.2,HEAT_ANOMALY
6	899,61.8,68.7,GAS_HEAT_EMERGENCY
7	996,59.2,69.9,GAS_HEAT_EMERGENCY
8	946,83.7,37.3,GAS_HEAT_EMERGENCY
9	961,50.9,77.3,GAS_DANGER
10	356,50.4,85.0,NORMAL
11	921,23.9,29.9,GAS_DANGER
12	902,80.5,28.1,GAS_HEAT_EMERGENCY
13	920,69.1,30.7,GAS_HEAT_EMERGENCY
14	885,47.7,53.6,GAS_DANGER
15	994,57.3,48.6,GAS_HEAT_EMERGENCY
16	214,56.5,80.3,HEAT_ANOMALY
17	376,43.0,29.0,NORMAL

kitchen_living_environmental_sa

Figure 23: Dataset table used for the environmental safety classifier.

The simulated gas level, temperature, and humidity were varied to produce examples of the five environmental safety states. Labels were generated automatically from the same safety definitions used in the embedded system, which avoided inconsistent manual annotation and ensured alignment between dataset and designed behaviour.

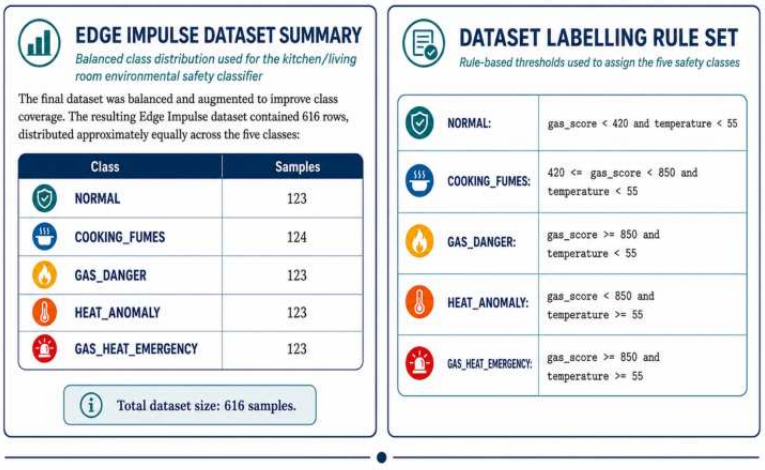


Figure 24: Edge Impulse dataset summary and rule-based labelling criteria for the kitchen/living room environmental safety classifier. The upper panel shows the final balanced class distribution across the five target classes, while the lower panel presents the threshold based rules used to assign labels from gas score and temperature conditions before model training.

Humidity was retained as a contextual feature even though the synthetic ground-truth labels are determined by gas score and temperature. The public dataset contains 616 synthetic rows across five approximately balanced classes. Any apparent humidity importance is therefore treated as correlation within this synthetic sample, not an independent causal signal. The surviving artefacts do not preserve the exact historical partition sizes, random seed, or `UNCERTAIN` configuration, so those settings are not reconstructed here.

4.4 Impulse Design

The Edge Impulse pipeline was configured as a small tabular classification impulse using the three numerical features `gas_score`, `temperature`, and `humidity`. The processing stage used raw data features with standard scaling, and feature-importance analysis was used to assess the contribution of each input.

The results matched the intended system logic: gas score was the most important feature, temperature the second most important, and humidity a smaller contextual feature. The learning block used a small neural-network classifier with three input features, two dense layers, and five output classes. This was appropriate because the dataset is small, numerical, and low-dimensional, so a larger model would have added complexity without clear deployment benefit.

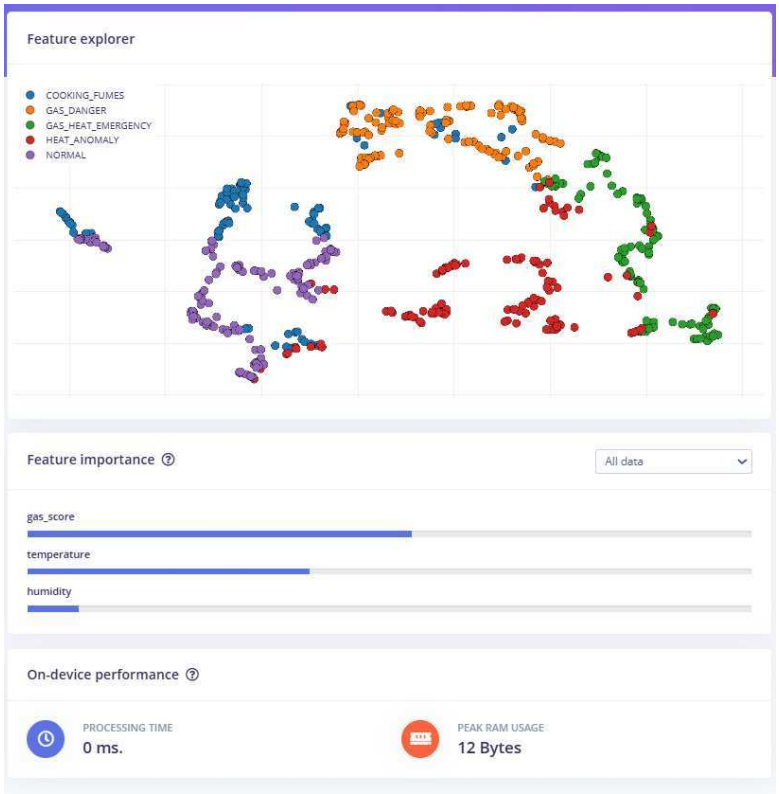


Figure 25: Edge Impulse outputs for the kitchen/living room environmental safety classifier. The upper part shows the feature explorer and the generated feature space for the five environmental safety classes. The lower part shows the feature importance analysis, where `gas_score` is the most influential feature, followed by `temperature` and `humidity`.

4.5 Training and Performance Metrics

The surviving Edge Impulse validation view reports 93.4% validation accuracy, an Edge Impulse-reported AUC of 1.00, weighted precision of 0.94, weighted recall of 0.93, and weighted F1 score of 0.93. A separate historical model-testing view reports 85.99% accuracy, an Edge Impulse-reported AUC of 0.99, weighted precision of 0.91, weighted recall of 0.89, and weighted F1 score of 0.89. Its confusion matrix also contains an `UNCERTAIN` outcome bucket. Because the original split sizes, seed, uncertainty threshold and full training configuration are not preserved, these are reported as historical platform outputs rather than independently recomputed metrics; ambiguity is concentrated mainly around `COOKING_FUMES` and `HEAT_ANOMALY`.



Figure 26: Historical Edge Impulse data explorer and platform-estimated on-device performance. The EON Compiler view reports approximately 1 ms inference time, 1.4 KB peak RAM, and 14.4 KB flash usage; these are platform estimates rather than measurements from the current Wokwi runtime.

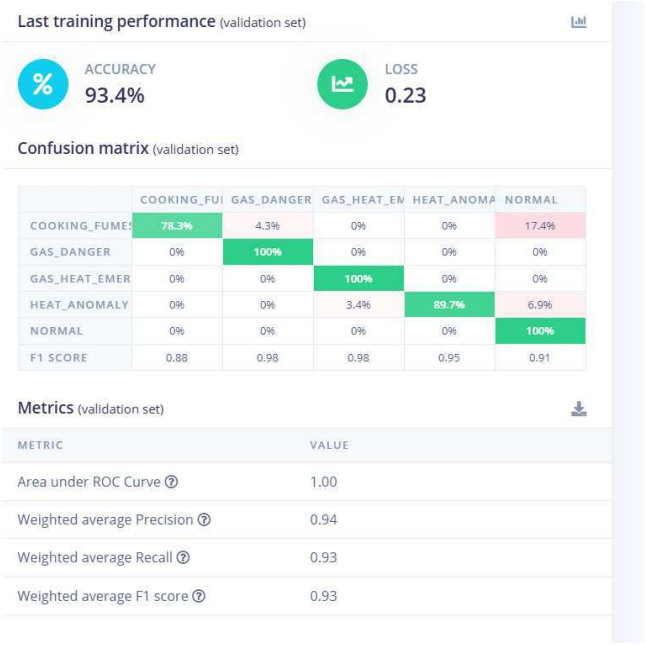


Figure 27: Historical Edge Impulse validation results. The validation set reports 93.4% accuracy, AUC 1.00, weighted precision 0.94, weighted recall 0.93, and weighted F1 score 0.93.

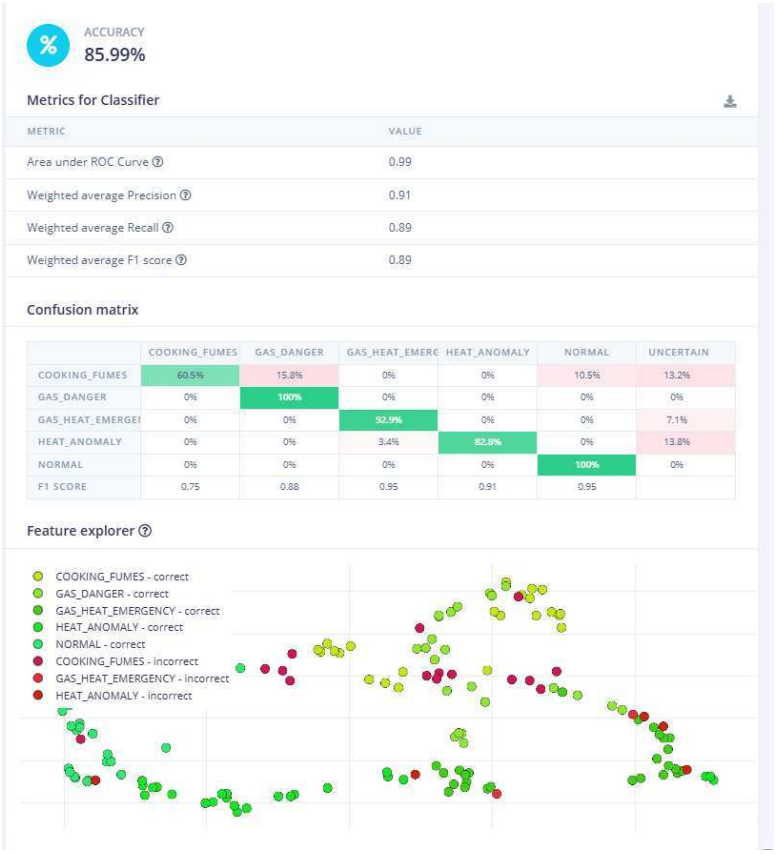


Figure 28: Historical Edge Impulse model-testing results. The platform reports 85.99% accuracy, AUC 0.99, weighted precision 0.91, weighted recall 0.89, and weighted F1 score 0.89. The `UNCERTAIN` column is a platform-reported outcome bucket; the historical threshold/configuration that produced it is not preserved in the public artefacts.

4.6 Deployment Strategy

The historical model was exported from Edge Impulse as an Arduino library using quantised int8 optimisation and the EON compiler. That generated archive is not present in the current public source tree; instead, the kitchen/living room code retains an optional inference path that can be enabled with a freshly generated library/header for the intended target (Edge Impulse, n.d.b). The target controller is the ESP32-S3 platform documented by Espressif Systems (n.d.).

In the hardened public Wokwi build, `USE_EDGE_IMPULSE_RUNTIME=0`. No ML inference is executed: actionable environmental state is deterministic and persistence-gated, ML confidence is `N/A`, and raw ML prediction is `DISABLED`. If genuine inference is enabled, its output remains advisory telemetry and does not drive blinds, AC, buzzer, or emergency-cooling actions.

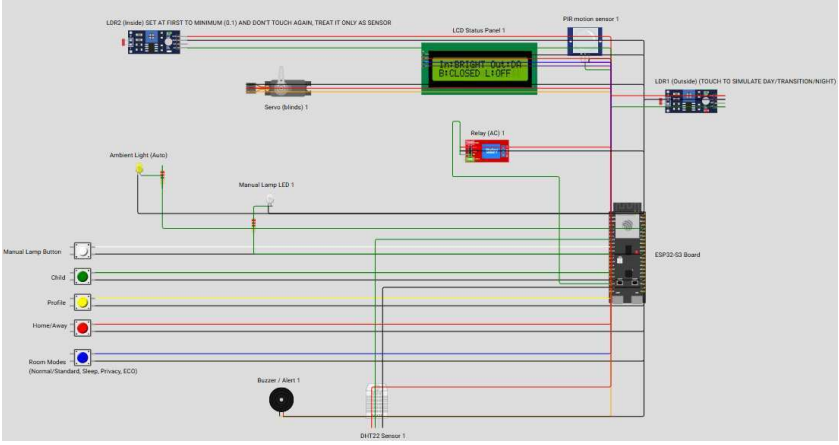
5 Implementation

5.1 Simulation Environment

The practical implementation was developed mainly in Wokwi. This allowed sensor integration, room logic, LCD outputs, serial monitoring, Blynk connectivity, and safety-related behaviour to be tested without full physical hardware assembly. Wokwi was especially useful for iterative debugging, threshold tuning, and controlled testing of gas, temperature, humidity, light, and motion changes (Wokwi, n.d.). The checked-in Wokwi library versions are pinned to the same versions used by GitHub Actions so browser-project dependencies and CI remain aligned.

5.2 Bedroom Module Implementation

The bedroom module was the most behaviourally detailed subsystem developed during the earlier stage of the project. It implements light-driven state classification, PIR-based occupancy detection with profile-dependent hold times, adaptive blinds, ambient support lighting, manual lamp integration, AC control through DHT22 readings, and local feedback through the LCD and Serial Monitor. If a DHT22 read becomes invalid, the hardened runtime forces the AC off rather than continuing to act on stale thermal data. It also supports child-mode protection, Away-mode behaviour, and authenticated state changes.



(a) Wokwi components and circuit connections.

```
======== SYSTEM STATUS (refresh every 5 seconds) =========
Room mode: NORMAL
Profile: STANDARD
House status: HOME
Pet at home: NO
Alert: NONE
AWAY motion count: 0
LDR2 inside raw: 2205 | inside effective brightness score: 711
LDR1 outside raw: 4063 | active outside score: 0 | source: BLYNK_V3
Inside state: BRIGHT
Outside state: NIGHT
PIR raw: 0
Presence active: YES
Manual lamp: ON
Blinds: CLOSED
Smart ambient light: OFF
Temperature: 60.7 C
Humidity: 100.0 %
Thermal state: WARM
AC: ON
Request active: NONE
```

(b) Serial Monitor outputs.

Figure 29: Bedroom implementation in Wokwi, showing the components and the Serial Monitor outputs.

5.3 Kitchen/Living Room Implementation

The kitchen/living room implementation follows the same broad software structure but adapts it to different priorities. Its main additions are gas sensing, persistence-gated emergency escalation, and an optional advisory Edge Impulse inference path. The room keeps a manual main lamp instead of automated ambient support lighting; actionable environmental state is deterministic in the hardened runtime.

The implementation includes DHT22 thermal sensing, MQ2 gas-score conversion, PIR motion detection, LDR-based brightness interpretation, servo blind control, AC relay behaviour, buzzer alerts, sprinkler or emergency-cooling LED indication, LCD output, Blynk monitoring, and the same Home/Away security logic used elsewhere in the house. Gas actions are gated by a 15 s warm-up, 2.5 s fumes persistence, 8 s normal danger persistence, 1.2 s severe-path persistence, and 5 s clear persistence; these are demonstration timings, not calibrated safety limits. Invalid DHT data forces AC off while gas monitoring remains available. The environmental-state pipeline is summarised below.

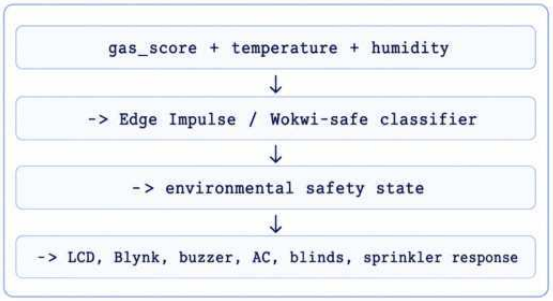
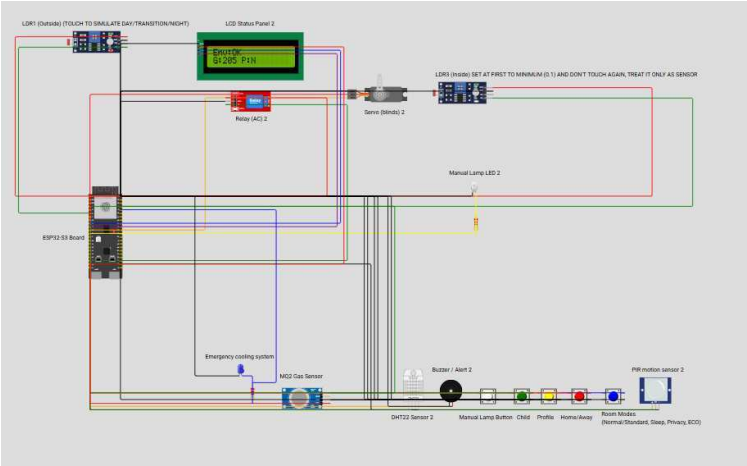


Figure 30: Historical development pipeline. In the hardened public Wokwi build, ML-disabled operation is rule-derived rather than simulated inference.

Simulation confirmed the deterministic safety behaviour. For example, a sustained gas score around 885 with temperature at 72.4 °C produced `GAS_HEAT_EMERGENCY`, activated the emergency-cooling state, opened the blinds, turned off the AC, and displayed the alert as *GAS + HIGH TEMP*. When gas later decreased while temperature remained high, the state moved to `HEAT_ANOMALY`, and when both fell below the project's demonstration thresholds, the status returned to `NORMAL`.



(a) Wokwi components and circuit connections.

```
======== KITCHEN/LIVING STATUS (refresh every 5 seconds) =========
Room mode: NORMAL
Profile: STANDARD
House status: HOME
Pet at home: NO
Alert: NONE
AWAY motion count: 0
LDR inside raw: 4063 | inside effective brightness score: 432
LDR outside raw: 610 | active outside score: 851 | source: WOKWI_LDR
Inside state: BRIGHT
Outside state: DAY
PIR raw: 0
Presence active: NO
Manual main lamp: OFF
Blinds: OPEN
MQ2 raw: 843 | gas score: 205
MQ2 warming up: NO
Gas thresholds - clear/warn/danger/severe: 350 / 420 / 850 / 970
Gas status: NORMAL
AI/environment status: NORMAL
Last ML prediction: NORMAL
ML valid: YES
ML confidence: 0.990
Emergency cooling threshold: 55.0 C
Emergency cooling active: NO
Temperature: 18.3 C
Humidity: 40.0 %
Thermal state: COOL
AC: OFF
Request active: NONE
AI environmental status (Wokwi-safe): NORMAL confidence=0.990
```

(b) Serial Monitor outputs.

Figure 31: Historical Wokwi implementation screenshot. The visible simulated ML-confidence output predates hardening; current ML-disabled output is `N/A`/`DISABLED`.

5.4 Blynk Mobile Dashboards and Remote Interaction

Blynk is optional in the hardened public build: `ENABLE_BLYNK=0` by default lets Wokwi run without private cloud credentials, while users can enable their own configuration explicitly.

The bedroom dashboard can display sensor and room states. The physical outdoor LDR is authoritative by default; if Blynk is enabled, V3 can take control until the LDR changes again.

The kitchen/living dashboard emphasises gas, thermal and emergency states. V29 reports the deterministic persistence-gated environmental status; V30/V31 show genuine advisory inference confidence/prediction only when enabled, otherwise `N/A`/`DISABLED` (Blynk, n.d.). Control widgets are periodically resynchronised with local state.

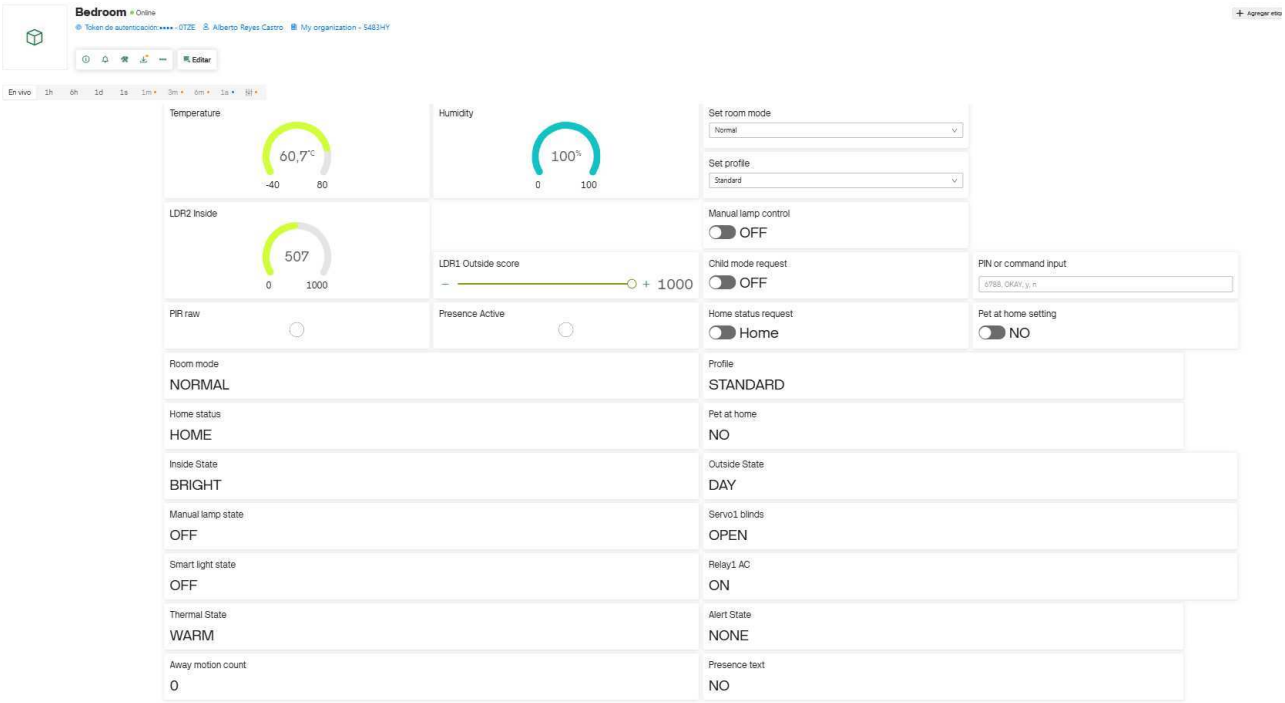


Figure 32: Bedroom Blynk dashboard used during development.

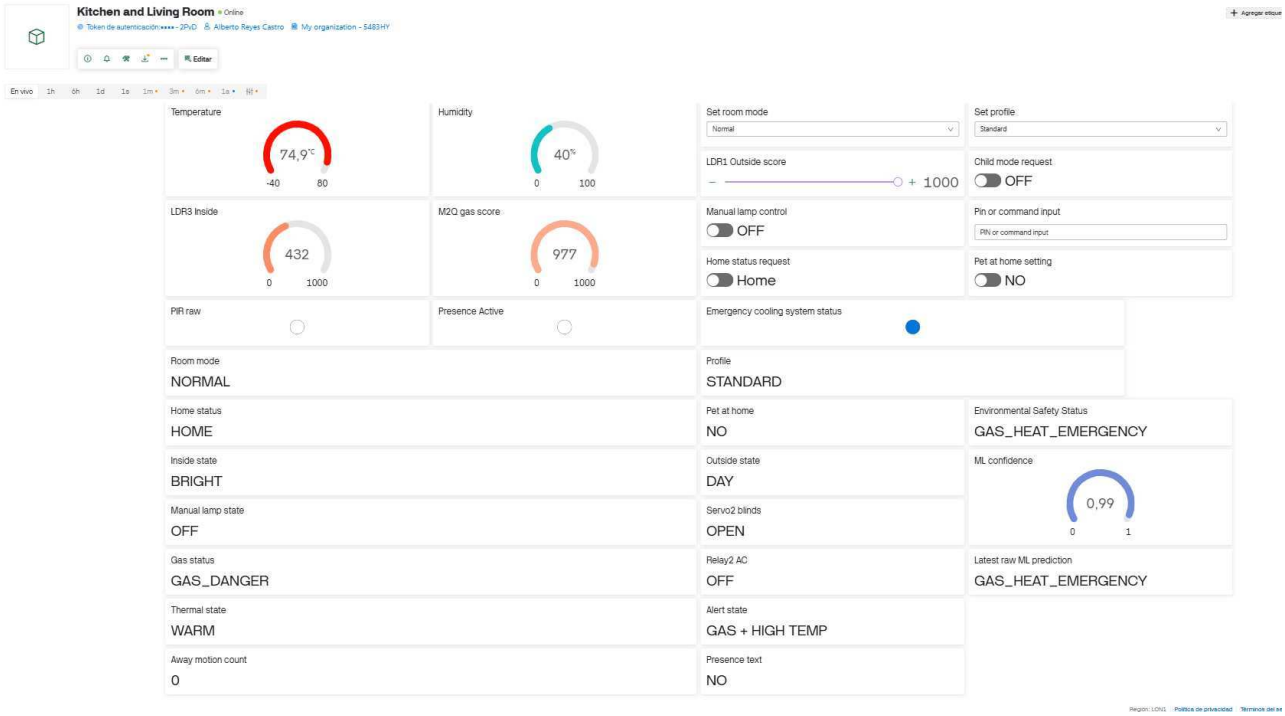


Figure 33: Historical kitchen/living Blynk dashboard from development. Any visible model/confidence values predate hardening; the current ML-disabled public build reports `N/A`/`DISABLED`.

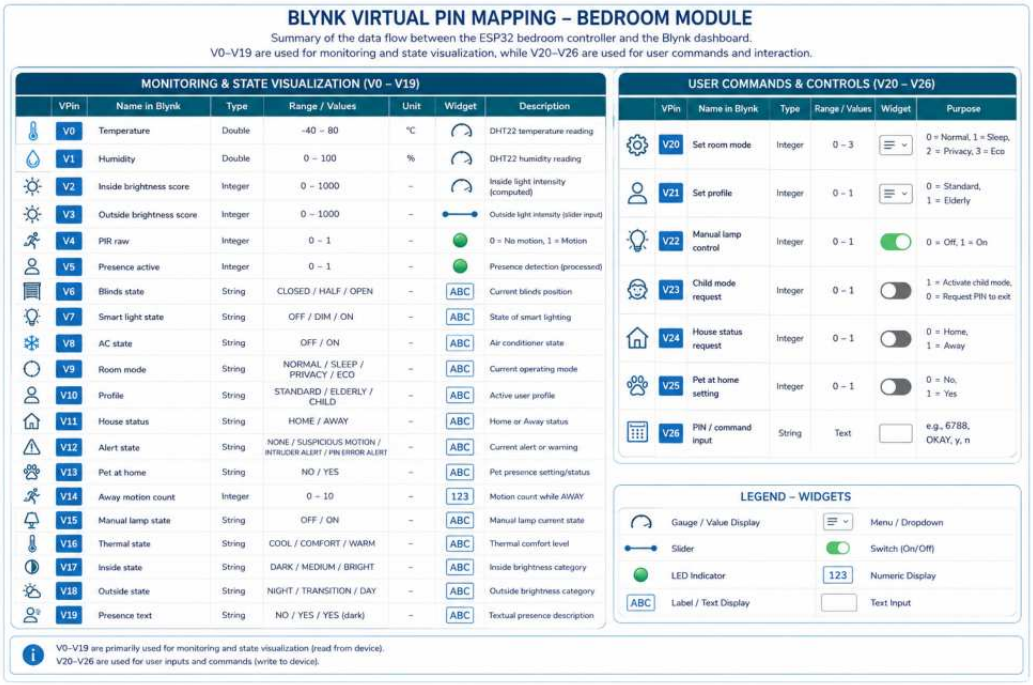


Figure 34: Bedroom Blynk virtual-pin mapping used during development. The historical graphic shows an earlier demonstration PIN example; the current public code uses demo PIN `1234` and secret word `OKAY`.

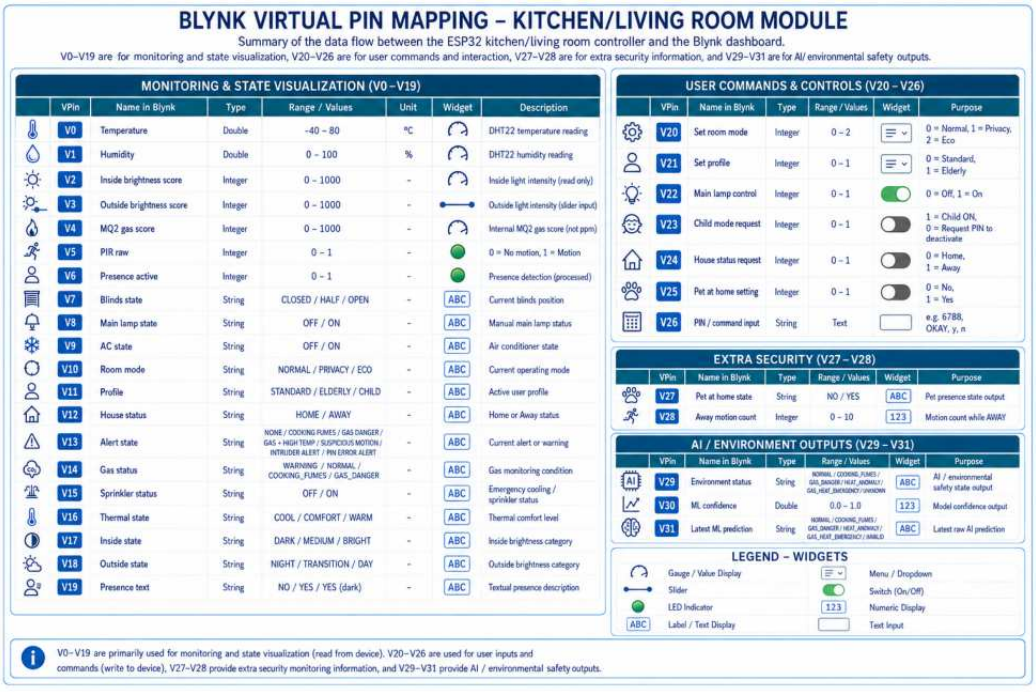


Figure 35: Kitchen/living Blynk virtual-pin mapping. V29 is environmental status; V30/V31 are genuine ML confidence/prediction when enabled, otherwise `N/A`/`DISABLED`.

5.5 Authentication and Security Logic

A recurring feature across the implementation is local authentication for sensitive state changes. Child-mode deactivation and Away-to-Home transition require authentication through the serial interface or Blynk command

input, while intruder-alert states use a separate secret-word mechanism for controlled clearance. This provides demonstration access-control safeguards for sensitive state changes; it is not production-grade authentication, but it helps prevent accidental reversal of protected room states.

6 Evaluation

The evaluation is based on scenario testing, simulation behaviour, build verification, and historical Edge Impulse metrics. The aim is not to present a finished commercial product but to assess whether each room behaves coherently under changing environmental and user conditions, whether temporal safety contracts hold, and whether the historical classifier evidence supports the documented TinyML workflow.

In the bedroom, the evaluation focused on night-time behaviour, profile-sensitive support logic, occupancy timing, blind transitions, thermal response, and Away-mode alert escalation. Results show that the same room can respond differently under *Standard*, *Elderly*, and *Child* profiles while maintaining a coherent overall behaviour.

In the kitchen/living room, the evaluation focused on gas-reading behaviour, persistence-based safety escalation, DHT failure handling, AC interaction, emergency-cooling activation, and deterministic environmental-state transitions; historical Edge Impulse metrics are reported separately. Scenario contracts verify that isolated gas spikes do not trigger actionable state, while the 2.5 s/8 s/1.2 s escalation windows and 5 s clear window are enforced. Simulation confirmed the intended behaviour: sustained high gas and temperature produced `GAS_HEAT_EMERGENCY`, activated the *GAS + HIGH TEMP* alert, opened the blinds, turned off the AC, and triggered emergency cooling; when gas fell but temperature remained high, the state moved to `HEAT_ANOMALY`; when both fell below the project's demonstration thresholds, the system returned to `NORMAL`.

Historical Edge Impulse validation reports 93.4% validation accuracy (AUC 1.00; weighted precision/recall/F1 = 0.94/0.93/0.93). The separate model-testing view reports 85.99% accuracy (AUC 0.99; weighted precision/recall/F1 = 0.91/0.89/0.89) and includes an `UNCERTAIN` outcome bucket whose historical threshold is not preserved. Edge Impulse also reports estimated ESP32-S3 performance of approximately 1 ms inference time, 1.4 KB peak RAM, and 14.4 KB flash usage; these remain platform estimates rather than current Wokwi measurements.
Repository-level verification complements the screenshots: `validate_repo.py` checks dataset, diagrams, public-surface, credential placeholders, pin mappings, dependency parity, and hardened runtime contracts; `test_scenarios.py` tests gas persistence/clearing, profile thresholds, deterministic environmental state, and DHT/AC fail-safe behaviour. GitHub Actions compiles both public ESP32-S3 sketches with pinned dependencies and separately compiles the optional Edge Impulse integration surface against a CI-only mock header. These are deterministic design/build checks, not substitutes for real-hardware validation.

A key strength of the prototype is that it demonstrates differentiated room logic within a shared architectural framework while also implementing an initial Edge Impulse workflow for environmental safety classification. The main limitation is that the dataset is simulated and rule-labelled rather than collected from physical sensor deployments, so real-world validation would still be required before treating the model as a safety-certified system.

7 Ethical, Privacy & Security Considerations

A main motivation for the project is privacy-aware local decision making. By keeping the main room logic on the embedded controller, the system avoids unnecessary transmission of household behavioural data to external services. This does not remove all privacy concerns, especially because Blynk introduces a cloud-connected dashboard layer, but it reduces the default exposure of raw indoor activity data. This follows a GDPR-style data-minimisation approach: only the information needed for room-state interpretation is processed or displayed, and the main decision logic remains local rather than continuously exporting raw household activity data (Information Commissioner's Office, n.d.).

The prototype includes demonstration access-control safeguards through protected state transitions, Away-mode

alert escalation, local alarm mechanisms, and room-level modularity. These mechanisms are not production-grade security; a real deployment would require secure credential provisioning, stronger authentication, transport security, update policy and threat modelling.

Ethically, the project recognises the limits of low-cost sensors and small machine-learning models. This is especially important for gas and safety-related sensing, so the system interprets gas readings carefully through staged escalation rather than claiming chemically precise identification. In the same way, the Edge Impulse classifier is treated as a supportive environmental classifier rather than a certified safety device.

The architecture is ethically stronger than an opaque AI-only approach because deterministic, persistence-gated safety rules remain authoritative. Even if genuine inference is enabled, its output is advisory telemetry and does not control safety actuators. With ML disabled, the public build reports that explicitly.

8 Discussion

This project shows that a smart home can be designed as a set of room-level embedded systems rather than as one undifferentiated automation layer. This is useful conceptually because each room can express its own priorities, and practically because each module can be implemented, tested, and refined separately.

The contrast between the bedroom and the kitchen/living room is one of the main strengths of the design. The bedroom demonstrates profile-sensitive behaviour around timing, support lighting, rest, and privacy, while the kitchen/living room is shaped more strongly by functional comfort, environmental safety, and emergency response.

The Edge Impulse workflow also clarifies the role of AI in the project. The model is not used as a decorative addition or as a replacement for all rules. Instead, it is applied to a specific problem where multi-sensor classification is useful: environmental safety interpretation in the kitchen/living room. Because the dataset was created in simulation and labelled using the project's own safety definitions, the model mainly learns the environmental boundaries designed during development. Its value is therefore not that it outperforms the original rules, but that it demonstrates a complete embedded machine-learning workflow and prepares the system for future real-sensor data. The resulting architecture combines explainable deterministic control with an optional learned-inference path for advisory telemetry; the hardened public default executes no ML inference.

At the same time, the prototype remains a staged implementation rather than a finished commercial product. Gas logic still requires real-world calibration and the dataset is simulated and rule-labelled. The hardened Wokwi build therefore disables Edge Impulse inference rather than emulating model output; genuine inference remains future target-hardware work.

9 Conclusion & Future Work

This report has presented a modular AIoT smart-home system based on ESP32-S3 room controllers. The project has shown how comfort, privacy, safety, and security can be treated as room-specific behaviours shaped by environment, user context, local control logic, remote monitoring, and lightweight embedded AI.

The bedroom and kitchen/living room modules show that a shared house-wide philosophy can coexist with different room priorities. The bedroom focuses on comfort, privacy, rest, and support for vulnerable users, while the kitchen/living room focuses on shared-space functionality, gas monitoring, heat anomaly detection, and emergency response.

A significant outcome is the historical Edge Impulse environmental safety classifier for the kitchen/living room. The recorded model classifies five trained states using gas score, temperature, and humidity; the historical Edge Impulse model-testing view reports 85.99% accuracy, with an additional `UNCERTAIN` outcome bucket in that

interface. The public implementation preserves an optional advisory integration path while keeping deterministic rules authoritative for all safety actions.

Future work should focus on five areas: real ESP32-S3 hardware testing with a freshly generated Edge Impulse Arduino library; expansion of the dataset with real MQ2 and DHT22 readings; calibration of gas thresholds and persistence timings using physical hardware; extension of the model with `COLD_ANOMALY` detection and heater response; and further refinement of the optional Blynk dashboards to improve user feedback, alert history, remote authentication flow, and genuine AI-related outputs when inference is enabled.

The broader value of the project lies in its architecture. Rather than treating AI as a decorative add-on to conventional automation, the project develops an edge-oriented smart home in which local reasoning is already central, making embedded AI both technically meaningful and conceptually justified.

References

Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M. and Ayyash, M. (2015) 'Internet of Things: A survey on enabling technologies, protocols, and applications', *IEEE Communications Surveys & Tutorials*, 17(4), pp. 2347–2376. DOI: 10.1109/COMST.2015.2444095.

Aosong Electronics (n.d.) *AM2302 Technical Manual: Temperature and Humidity Sensor*. Available at: https://www.aosong.com/uploadfiles/2025/04/20250417105409216.pdf (Accessed: 24 October 2026).

Blynk (n.d.) *Blynk Documentation*. Available at: https://docs.blynk.io/ (Accessed: 24 October 2026).

Dutta, L. and Bharali, S. (2021) 'TinyML meets IoT: A comprehensive survey', *Internet of Things*, 16, Article 100461. DOI: 10.1016/j.iot.2021.100461.

Edge Impulse (n.d.a) *Edge Impulse Documentation*. Available at: https://docs.edgeimpulse.com/ (Accessed: 24 October 2026).

Edge Impulse (n.d.b) *Run Arduino library (IDE 2.0)*. Available at: https://docs.edgeimpulse.com/hardware/deployments/run-arduino-2-0 (Accessed: 24 October 2026).

Espressif Systems (n.d.) *ESP32-S3 Series Datasheet*. Available at: https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf (Accessed: 24 October 2026).

Information Commissioner's Office (n.d.) *Principle (c): Data minimisation*. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-protection-principles/a-guide-to-the-data-protection-principles/data-minimisation/ (Accessed: 24 October 2026).

Merenda, M., Porcaro, C. and Iero, D. (2020) 'Edge machine learning for AI-enabled IoT devices: A review', *Sensors*, 20(9), Article 2533. DOI: 10.3390/s20092533.

Ray, P.P. (2018) 'A survey on Internet of Things architectures', *Journal of King Saud University – Computer and Information Sciences*, 30(3), pp. 291–319. DOI: 10.1016/j.jksuci.2016.10.003.

Shi, W., Cao, J., Zhang, Q., Li, Y. and Xu, L. (2016) 'Edge computing: Vision and challenges', *IEEE Internet of Things Journal*, 3(5), pp. 637–646. DOI: 10.1109/JIOT.2016.2579198.

Singh, R. and Gill, S.S. (2023) 'Edge AI: A survey', *Internet of Things and Cyber-Physical Systems*, 3, pp. 71–92. DOI: 10.1016/j.iotcps.2023.02.004.

Stojkoska, B.L.R. and Trivodaliev, K.V. (2017) 'A review of Internet of Things for smart home: Challenges and solutions', *Journal of Cleaner Production*, 140, pp. 1454–1464. DOI: 10.1016/j.jclepro.2016.10.006.

Warden, P. and Situnayake, D. (2019) *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. Sebastopol, CA: O'Reilly Media.

Wokwi (n.d.) *Wokwi Documentation*. Available at: https://docs.wokwi.com/ (Accessed: 24 October 2026).

Zhengzhou Winsen Electronics Technology Co., Ltd. (2021) *MQ-2 Manual*. Available at: https://www.winsen-sensor.com/d/files/newpdf/mq-2-%28ver1_6%29---manual.pdf (Accessed: 24 October 2026).

A Appendix A: Additional Simulation Figures (Wokwi)

This appendix includes a small selection of Wokwi screenshots rather than every possible scenario (as there are many of them). The most useful examples are an Away mode alarm, gas alert, and activated smart lights (when inside state is dark, manual lamp is off and motion detected in the dark), as these demonstrate the main behaviours without overloading the appendix.

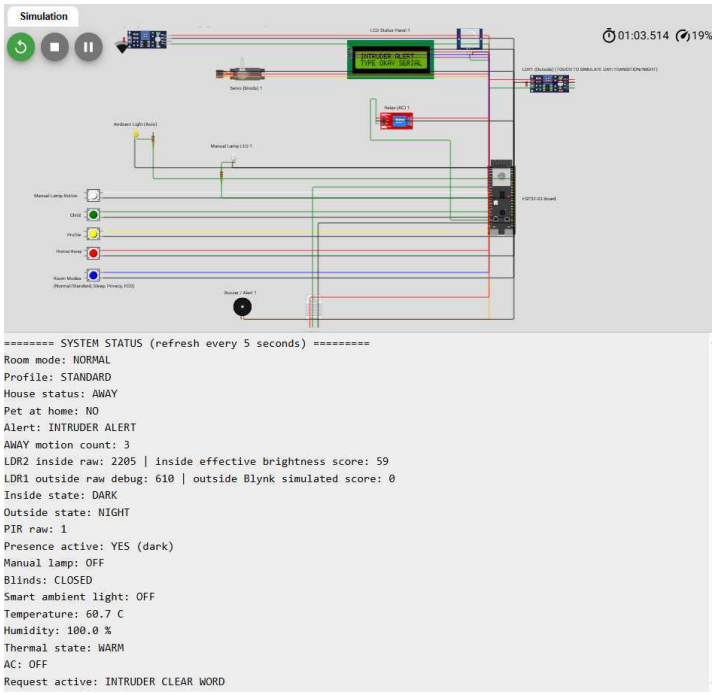


Figure 36: Intruder Alert (bedroom)

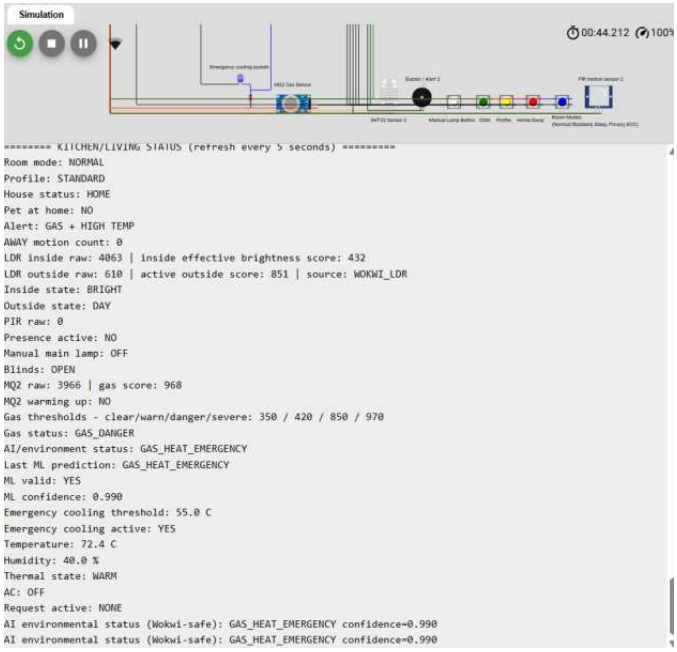


Figure 37: Gas-and-high-temperature emergency cooling (historical pre-hardening screenshot; current ML-disabled output is `N/A/DISABLED`).

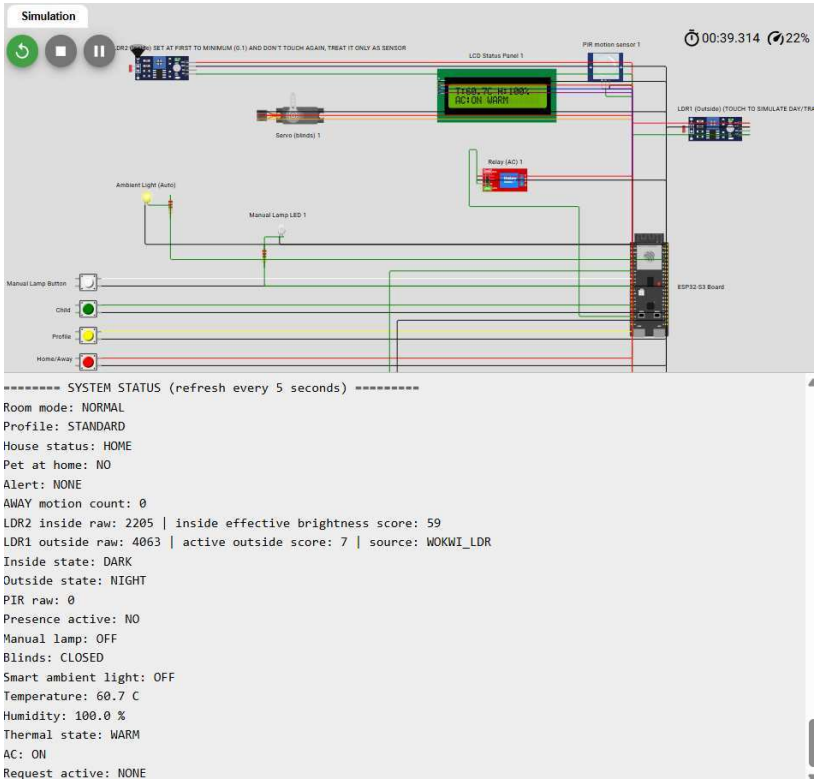


Figure 38: Smart lights ON

B Appendix B: Public Repository and Supporting Material

Public code, dataset documentation, report material and supporting assets are available here:

https://github.com/albertoreyescastro/AIoT-smart-home-esp32-wokwi.

C Appendix C: Brief User Guide

C.1 Purpose

This modular ESP32-S3 prototype contains bedroom and kitchen/living-room controllers with local sensing/automation, optional Blynk monitoring, and an optional integration path for the historical Edge Impulse classifier.

C.2 Required Platforms and Materials

To reproduce the current prototype, the user should have:

- the authoritative checked-in Wokwi project files and Arduino/ESP32-S3 source,
- optionally, personal Blynk credentials/dashboard configuration,
- optionally, a freshly generated Edge Impulse Arduino library for genuine inference, plus the public dataset/supporting assets.

Running the Bedroom Module

1. Open the bedroom Wokwi project.
2. Start the simulation and use the physical buttons to change room mode, profile, house state, manual lamp state, and child mode; Blynk controls are available only when explicitly enabled.
3. The physical outdoor LDR is the default outside-light source. If Blynk is enabled, V3 can take control until the physical LDR changes again.

4. Observe the resulting changes in blinds, smart lighting, AC behaviour, LCD output, Serial Monitor, and Blynk dashboard.

Running the Kitchen/Living Room Module

1. Open the kitchen/living room Wokwi project, start the simulation, and wait for the initial MQ2 warm-up before interpreting gas-related states.
2. Adjust gas, temperature, humidity, motion, and brightness conditions in the simulator.
3. As in the bedroom module, the physical outdoor LDR is the default source; an enabled Blynk V3 control can take over until the physical LDR changes again.
4. Observe the resulting environmental safety state, gas alerts, AC behaviour, blinds response, buzzer activation, emergency cooling indicator, LCD output, Serial Monitor, and Blynk dashboard.

Important Simulation Notes

- The **physical outdoor LDR is authoritative by default**; enabled Blynk V3 control can temporarily take over until the physical LDR changes again.
- **Blynk is disabled by default**; the local simulation therefore needs no private cloud credentials.
- The **indoor LDR remains active**; for repeatable demonstrations it can be left at a stable low reference value.
- **Sleep mode exists only in the bedroom**; the kitchen/living room uses Normal, Privacy and Eco.
- The kitchen/living room **MQ2 warm-up period is 15 seconds**; gas-related states should not be interpreted before it finishes.
- If a DHT read is invalid, the hardened code forces the **AC off**; gas monitoring remains available independently.

Blynk Interaction

Blynk is optional and disabled by default. With personal credentials and `ENABLE_BLYNK=1`, dashboards can expose supported monitoring/control functions; bedroom V3 may temporarily provide outside-light input, while kitchen/living V29 reports deterministic persistence-gated environmental status and V30/V31 report genuine advisory inference outputs only, otherwise `N/A`/`DISABLED`.

Edge Impulse / AI Operation

The kitchen/living room combines persistence-gated deterministic gas escalation with an optional classifier using gas score, temperature, and humidity. In the hardened public Wokwi build, `USE_EDGE_IMPULSE_RUNTIME=0`; genuine inference requires a freshly generated library/header and explicit enablement, and remains advisory telemetry rather than an actuator-control path.

Main Alerts and Responses

The most important expected system responses are:

- **SUSPICIOUS MOTION / INTRUDER ALERT**: triggered in Away mode after repeated motion detection.
- **COOKING_FUMES**: project-defined intermediate gas-score state; the label is synthetic and does not chemically identify cooking fumes.
- **GAS_DANGER**: project-defined high gas-score condition that triggers a defensive response.
- **HEAT_ANOMALY**: temperature above the project threshold without a simultaneous high gas-score condition.
- **GAS_HEAT_EMERGENCY**: high gas score combined with temperature above the project threshold; activates emergency-cooling indication and defensive room behaviour.

Authentication and Protected Actions

Some state changes require authentication:

- **Child mode deactivation** requires the system PIN (**1234**)
- **Away to Home transition** requires the system PIN (**1234**).
- **Intruder alert clearance** requires the secret word (**OKAY**).

Commands can be entered through Serial Monitor and, where supported, Blynk. The PIN/secret word are demonstration values only; any real deployment would require proper authentication and credential provisioning rather than hard-coded shared secrets.

Limitations. Simulation only; ML evidence is historical, safety actions are deterministic and persistence-gated, and invalid DHT data forces AC off. Real-sensor/hardware validation remains future work.